Applied Partial Differential Equations with Fourier Series and Boundary Value Problems

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All errors, typographical and substantive, and other offenses, are entirely my own.

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2 - Method of Separation of Variables

2.2 - Linearity

Question: 2.2.1

Show that any linear combination of linear operators is a linear operator.

Let c_1 and c_2 be constants and u_1 and u_2 be functions. Define

$$L(c_1u_1 + c_2u_2) = \sum_{i=1}^{n} d_i L_i(c_1u_1 + c_2u_2)$$

where the L_i's are each linear operators and the d_i's are constants. By the definition of linear operator, we can demonstrate

$$\begin{aligned} L(c_1u_1 + c_2u_2) &= \sum_{i=1}^{n} d_i L_i(c_1u_1 + c_2u_2) \\ &= \sum_{i=1}^{n} d_i [c_1 L_i(u_1) + c_2 L_i(u_2)] \\ &= c_1 \sum_{i=1}^{n} d_i L_i(u_1) + c_2 \sum_{i=1}^{n} d_i L_i(u_2) \end{aligned}$$

Thereby establishing that L is also a linear operator.

Question: 2.2.2

(a) Show that $L(u) = \frac{\partial}{\partial x}\left[K_0(x)\,\frac{\partial u}{\partial x}\right]$ is a linear operator.

(b) Show that usually $L(u) = \frac{\partial}{\partial x}\left[K_0(x,u)\,\frac{\partial u}{\partial x}\right]$ is not a linear operator.

Let c_1 and c_2 be constants and u_1 and u_2 be functions.

(a)

$$\begin{aligned} L(c_1u_1 + c_2u_2) &= \frac{\partial}{\partial x}\left[K_0(x)\,\frac{\partial}{\partial x}(c_1u_1 + c_2u_2)\right] \\ &= \frac{\partial}{\partial x}\left[K_0(x)\left(c_1\frac{\partial u_1}{\partial x} + c_2\frac{\partial u_2}{\partial x}\right)\right] \\ &= c_1\frac{\partial}{\partial x}\left[K_0(x)\,\frac{\partial u_1}{\partial x}\right] + c_2\frac{\partial}{\partial x}\left[K_0(x)\,\frac{\partial u_2}{\partial x}\right] \\ &= c_1L(u_1) + c_2L(u_2) \end{aligned}$$

(b) As given, one would need to demonstrate

$$\begin{aligned} L(c_1u_1 + c_2u_2) &= \frac{\partial}{\partial x}\left[K_0(x, c_1u_1 + c_2u_2)\,\frac{\partial}{\partial x}(c_1u_1 + c_2u_2)\right] \\ &= c_1\frac{\partial}{\partial x}\left[K_0(x,u_1)\,\frac{\partial u_1}{\partial x}\right] + c_2\left[K_0(x,u_2)\,\frac{\partial u_2}{\partial x}\right] \end{aligned}$$

And for the sake of argument, suppose

$$K_0(x, c_1u_1 + c_2u_2) = c_1K_0(x, u_1) + c_2K_0(x, u_2)$$

Then we would have

$$\begin{aligned}
L(c_1u_1 + c_2u_2) &= \frac{\partial}{\partial x}\left[K_0(x, c_1u_1 + c_2u_2)\,\frac{\partial}{\partial x}(c_1u_1 + c_2u_2)\right] \\
&= \frac{\partial}{\partial x}\left[(c_1K_0(x, u_1) + c_2K_0(x, u_2))\left(c_1\frac{\partial u_1}{\partial x} + c_2\frac{\partial u_2}{\partial x}\right)\right] \\
&= c_1^2\frac{\partial}{\partial x}\left[K_0(x, u_1)\,\frac{\partial u_1}{\partial x}\right] + c_2^2\left[K_0(x, u_2)\,\frac{\partial u_2}{\partial x}\right] \\
&\qquad + c_1c_2\frac{\partial}{\partial x}\left[K_0(x, u_1)\,\frac{\partial u_2}{\partial x}\right] + c_1c_2\left[K_0(x, u_2)\,\frac{\partial u_1}{\partial x}\right]
\end{aligned}$$

But even here, the cross-terms must be guaranteed to vanish, and the squared constant terms do not adhere to the definition of linearity. Under the given information, there is no way to determine if $K_0(x, u)$ decomposes in the way that it needs to enable this second equality to be true.

Question: 2.2.3

Show that $\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2} + Q(u, x, t)$ is linear if $Q = \alpha(x, t)\,u + \beta(x, t)$ and, in addition, homogeneous if $\beta(x, t) = 0$.

Define

$$L(u) = \frac{\partial u}{\partial t} - k\frac{\partial^2 u}{\partial x^2} - \alpha(x, t)\,u = \beta(x, t)$$

To test for linearity, compute

$$\begin{aligned}
L(c_1u_1 + c_2u_2) &= \frac{\partial}{\partial t}(c_1u_1 + c_2u_2) - k\frac{\partial^2}{\partial x^2}(c_1u_1 + c_2u_2) \\
&\quad - \alpha(x, t)(c_1u_1 + c_2u_2) \\
&= c_1\frac{\partial u_1}{\partial t} + c_2\frac{\partial u_2}{\partial t} - c_1k\frac{\partial^2 u_1}{\partial x^2} - c_2k\frac{\partial^2 u_2}{\partial x^2} \\
&\quad - c_1\alpha(x, t)\,u_1 - c_2\alpha(x, t)\,u_2 \\
&= c_1L(u_1) + c_2L(u_2)
\end{aligned}$$

By definition of homogeneity, $\beta(x, t) = 0$ produces the homogeneous condition.

Question: 2.2.4

In this exercise we derive superposition principles for nonhomogeneous problems.

(a) Consider $L(u) = f$. If u_p is a particular solution, $L(u_p) = f$, and if u_1 and u_2 are homogeneous solutions, $L(u_i) = 0$, show that $u = u_p + c_1u_1 + c_2u_2$ is another particular solution.

(b) If $L(u) = f_1 + f_2$, where u_{pi} is a particular solution corresponding to f_i, what is a particular solution for $f_1 + f_2$?

(a) $L(u_p + c_1u_1 + c_2u_2) = L(u_p) + c_1L(u_1) + c_2L(u_2) = f + 0 + 0 = f$

(b) One particular solution is $u = u_{p1} + u_{p2}$. In general, we can have

$$u = c_1u_1 + c_2u_2 + u_{p1} + u_{p2}$$

Question: 2.2.5

If L is a linear operator, show that $L\left(\sum_{n=1}^{M} c_n u_n\right) = \sum_{n=1}^{M} c_n L(u_n)$. Use this result to show that the principle of superposition may be extended to any finite number of homogeneous solutions.

Compute from the definition of linearity

$$L\left(\sum_{n=1}^{M} c_n u_n\right) = \sum_{n=1}^{M} L(c_n u_n) = \sum_{n=1}^{M} c_n L(u_n)$$

Assuming homogeneity for every solution, we can do (pedantically) induction on M to demonstrate that the result holds for any finite natural number.

$$L\left(\sum_{n=1}^{M} c_n u_n + c_{M+1}u_{M+1}\right) = \sum_{n=1}^{M} L(c_n u_n) + L(c_{M+1}u_{M+1}) = \sum_{n=1}^{M+1} c_n L(u_n)$$

2.3 - Heat Equation with Zero Temperatures at Finite Ends

Question: 2.3.1

For the following partial differential equations, what ordinary differential equations are implied by the method of separation of variables?

(a) $\dfrac{\partial u}{\partial t} = \dfrac{k}{r}\dfrac{\partial}{\partial r}\left(r\dfrac{\partial u}{\partial r}\right)$

(b) $\dfrac{\partial u}{\partial t} = k\dfrac{\partial^2 u}{\partial x^2} - v_0\dfrac{\partial u}{\partial x}$

(c) $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial u}{\partial y} = 0$

(d) $\dfrac{\partial u}{\partial t} = \dfrac{k}{r^2}\dfrac{\partial}{\partial r}\left(r^2\dfrac{\partial u}{\partial r}\right)$

(e) $\dfrac{\partial u}{\partial t} = k\dfrac{\partial^4 u}{\partial x^4}$

(f) $\dfrac{\partial^2 u}{\partial t^2} = c^2\dfrac{\partial^2 u}{\partial x^2}$

(a) Let $u(r,t) = \phi(r)\,G(t)$. Then

$$\begin{aligned}\frac{\partial u}{\partial t} &= \phi(r)\,\frac{dG}{dt}\\ \frac{k}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) &= \frac{k}{r}G(t)\,\frac{d}{dr}\left(r\frac{d\phi}{dr}\right)\end{aligned}$$

Setting equal and separating variables, we find

$$\begin{aligned}\phi(r)\,\frac{dG}{dt} &= \frac{k}{r}G(t)\,\frac{d}{dr}\left(r\frac{d\phi}{dr}\right)\\ \Longrightarrow\quad \frac{1}{kG(t)}\frac{dG}{dt} &= \frac{1}{r\phi(r)}\frac{d}{dr}\left(r\frac{d\phi}{dr}\right) = -\lambda\end{aligned}$$

This produces ordinary differential equations

$$\frac{dG}{dt} = -\lambda kG(t)\,, \qquad \frac{1}{r}\frac{d}{dr}\left(r\frac{d\phi}{dr}\right) = -\lambda\phi(r)$$

(b) Let $u(x,t) = \phi(x)\,G(t)$. Then

$$\begin{aligned}\frac{\partial u}{\partial t} &= \phi(x)\,\frac{dG}{dt}\\ k\frac{\partial^2 u}{\partial x^2} - v_0\frac{\partial u}{\partial x} &= G(t)\left[k\frac{d^2\phi}{dx^2} - v_0\frac{d\phi}{dx}\right]\\ &= kG(t)\left[\frac{d^2\phi}{dx^2} - \frac{v_0}{k}\frac{d\phi}{dx}\right]\end{aligned}$$

Setting equal and separating variables, we find

$$\begin{aligned} \phi(x)\,\frac{dG}{dt} &= kG(t)\left[\frac{d^2\phi}{dx^2} - \frac{v_0}{k}\frac{d\phi}{dx}\right] \\ \implies \frac{1}{kG(t)}\frac{dG}{dt} &= \frac{1}{\phi(x)}\left[\frac{d^2\phi}{dx^2} - \frac{v_0}{k}\frac{d\phi}{dx}\right] = -\lambda \end{aligned}$$

This produces ordinary differential equations

$$\frac{dG}{dt} = -\lambda k G(t)\,, \qquad \frac{d^2\phi}{dx^2} - \frac{v_0}{k}\frac{d\phi}{dx} = -\lambda\phi(x)$$

(c) Let $u(x,y) = \phi(x)\,G(y)$. Then

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial y} = \frac{d^2\phi}{dx^2}G(y) + \phi(x)\,\frac{dG}{dy} = 0$$

Setting equal to zero and separating variables, we find

$$\frac{1}{\phi(x)}\frac{d^2\phi}{dx^2} = -\frac{1}{G(y)}\frac{dG}{dy} = -\lambda$$

This produces ordinary differential equations

$$\frac{d^2\phi}{dx^2} = -\lambda\phi(x)\,, \qquad \frac{dG}{dy} = \lambda G(y)$$

(d) Let $u(r,t) = \phi(r)\,G(t)$. Then

$$\begin{aligned} \frac{\partial u}{\partial t} &= \phi(r)\,\frac{dG}{dt} \\ \frac{k}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right) &= G(t)\,\frac{k}{r^2}\frac{d}{dr}\left(r^2\frac{d\phi}{dr}\right) \end{aligned}$$

Setting equal and separating variables, we find

$$\begin{aligned} \phi(r)\,\frac{dG}{dt} &= G(t)\,\frac{k}{r^2}\frac{d}{dr}\left(r^2\frac{d\phi}{dr}\right) \\ \implies \frac{1}{kG(t)}\frac{dG}{dt} &= \frac{1}{r^2\phi(r)}\frac{d}{dr}\left(r^2\frac{d\phi}{dr}\right) = -\lambda \end{aligned}$$

This produces ordinary differential equations

$$\frac{dG}{dt} = -\lambda k G(t)\,, \quad \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\phi}{dr}\right) = -\lambda\phi(r)$$

(e) Let $u(x,t) = \phi(x)\,G(t)$. Then

$$\begin{aligned} \frac{\partial u}{\partial t} &= \phi(x)\,\frac{dG}{dt} \\ k\frac{\partial^4 u}{\partial x^4} &= kG(t)\,\frac{d^4\phi}{dx^4} \end{aligned}$$

Setting equal and separating variables, we find

$$\begin{aligned} \phi(x)\,\frac{dG}{dt} &= kG(t)\,\frac{d^4\phi}{dx^4} \\ \implies \frac{1}{kG(t)}\frac{dG}{dt} &= \frac{1}{\phi(x)}\frac{d^4\phi}{dx^4} = -\lambda \end{aligned}$$

This produces ordinary differential equations

$$\frac{dG}{dt} = -\lambda kG(t)\,, \quad \frac{d^4\phi}{dx^4} = -\lambda\phi(x)$$

(f) Let $u(x,t) = \phi(x)\,G(t)$. Then

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2} &= \phi(x)\,\frac{d^2G}{dt^2} \\ c^2\frac{\partial^2 u}{\partial x^2} &= c^2G(t)\,\frac{d^2\phi}{dx^2} \end{aligned}$$

Setting equal and separating variables, we find

$$\begin{aligned} \phi(x)\,\frac{d^2G}{dt^2} &= c^2G(t)\,\frac{d^2\phi}{dx^2} \\ \implies \frac{1}{c^2G(t)}\frac{d^2G}{dt^2} &= \frac{1}{\phi(x)}\frac{d^2\phi}{dx^2} = -\lambda \end{aligned}$$

This produces ordinary differential equations

$$\frac{d^2G}{dt^2} = -\lambda c^2G(t)\,, \quad \frac{d^2\phi}{dx^2} = -\lambda\phi(x)$$

Question: 2.3.2

Consider the differential equation

$$\frac{d^2\phi}{dx^2} + \lambda\phi = 0$$

Determine the eigenvalues λ (and corresponding eigenfunctions) if ϕ satisfies the following boundary conditions. Analyze three cases ($\lambda > 0$, $\lambda = 0$, $\lambda < 0$). You may assume that the eigenvalues are real.

(a) $\phi(0) = 0$ and $\phi(\pi) = 0$

(b) $\phi(0) = 0$ and $\phi(1) = 0$

(c) $\dfrac{d\phi}{dx}(0) = 0$ and $\dfrac{d\phi}{dx}(L) = 0$ (If necessary, see Section 2.4.1.)

(d) $\phi(0) = 0$ and $\dfrac{d\phi}{dx}(L) = 0$

(e) $\dfrac{d\phi}{dx}(0) = 0$ and $\phi(L) = 0$

(f) $\phi(a) = 0$ and $\phi(b) = 0$ (You may assume that $\lambda > 0$.)

(g) $\phi(0) = 0$ and $\dfrac{d\phi}{dx}(L) + \phi(L) = 0$ (If necessary, see Section 5.8.)

The general forms for each case are

$$\begin{aligned} &\phi = c_1 \cos\left(\sqrt{\lambda}x\right) + c_2 \sin\left(\sqrt{\lambda}x\right) && (\lambda > 0) \\ &\phi = c_1 + c_2 x && (\lambda = 0) \\ &\phi = c_1 \cosh\left(\sqrt{s}x\right) + c_2 \sinh\left(\sqrt{s}x\right) && (\lambda < 0, s = -\lambda) \end{aligned}$$

(a) $\underline{\lambda > 0}$ The boundary conditions produce

$$\begin{aligned} \phi(0) &= c_1 = 0 \\ \phi(\pi) &= c_2 \sin\left(\sqrt{\lambda}\pi\right) = 0 \end{aligned}$$

which forces $\sqrt{\lambda} = n$, implying $\lambda = n^2$ for positive integers. Then $\phi(x) = c_2 \sin(nx)$.

$\underline{\lambda = 0}$ The boundary conditions produce

$$\begin{aligned} \phi(0) &= c_1 = 0 \\ \phi(\pi) &= c_2 \pi = 0 \end{aligned}$$

implying $c_2 = 0$, so we only have the trivial solution, confirming that $\lambda = 0$ cannot be an eigenvalue.

$\underline{\lambda < 0}$ Let $-\lambda = s$. The boundary conditions produce

$$\begin{aligned} \phi(0) &= c_1 = 0 \\ \phi(\pi) &= c_2 \sinh\left(\sqrt{s}\pi\right) = 0 \end{aligned}$$

which requires either $s = 0$ or $c_2 = 0$. We consider only $s > 0$, so that forces $c_2 = 0$, and the implication that $\lambda < 0$ also cannot be an eigenvalue.

(b) $\underline{\lambda > 0}$ The boundary conditions produce

$$\begin{aligned}\phi(0) &= c_1 = 0 \\ \phi(1) &= c_2 \sin\left(\sqrt{\lambda}\right) = 0\end{aligned}$$

which forces $\lambda = (n\pi)^2$ for positive integers. Then $\phi(x) = c_2 \sin(n\pi x)$.

$\underline{\lambda = 0}$ The boundary conditions produce

$$\begin{aligned}\phi(0) &= c_1 = 0 \\ \phi(1) &= c_2 = 0\end{aligned}$$

leaving us only with the trivial solution, implying that $\lambda = 0$ cannot be an eigenvalue.

$\underline{\lambda < 0}$ Let $-\lambda = s$. The boundary conditions produce

$$\begin{aligned}\phi(0) &= c_1 = 0 \\ \phi(1) &= c_2 \sinh\left(\sqrt{s}\right) = 0\end{aligned}$$

which requires either $s = 0$ or $c_2 = 0$. We consider only $s > 0$ (equivalent with $\lambda < 0$), so we must have $c_2 = 0$, ergo $\lambda < 0$ also cannot be an eigenvalue.

(c) $\underline{\lambda > 0}$ The boundary conditions produce

$$\begin{aligned}\frac{d\phi}{dx}(0) &= c_2\sqrt{\lambda} = 0 \qquad \implies c_2 = 0 \\ \frac{d\phi}{dx}(L) &= -c_1\sqrt{\lambda}\sin\left(\sqrt{\lambda}L\right) = 0\end{aligned}$$

which requires $\lambda = (n\pi/L)^2$ for positive integers. Then $\phi(x) = c_1 \cos(n\pi x/L)$.

$\underline{\lambda = 0}$ The boundary conditions produce

$$\frac{d\phi}{dx}(0) = \frac{d\phi}{dx}(L) = c_2 = 0$$

Then $\phi(x) = c_1$, thus establishing $\lambda = 0$ as an eigenvalue for this function.

$\underline{\lambda < 0}$ Let $-\lambda = s$. The boundary conditions produce

$$\begin{aligned}\frac{d\phi}{dx}(0) &= c_2\sqrt{s} = 0 \qquad \implies c_2 = 0 \\ \frac{d\phi}{dx}(L) &= c_1\sqrt{s}\sinh\left(\sqrt{s}L\right) = 0\end{aligned}$$

which requires $c_1 = 0$ as well, as neither $s = 0$ nor $L = 0$ can be true. Therefore $\lambda < 0$ does not exist as an eigenvalue.

(d) $\underline{\lambda > 0}$ The boundary conditions produce

$$\phi(0) = c_1 = 0$$
$$\frac{d\phi}{dx}(L) = c_2\sqrt{\lambda}\cos\left(\sqrt{\lambda}L\right) = 0$$

which requires $\lambda = [(2n-1)\,\pi/2L]^2 = [(n-1/2)\,\pi/L]^2$ for positive integers. Then

$$\phi(x) = c_2 \sin\left(\frac{\left(n-\frac{1}{2}\right)\pi x}{L}\right)$$

$\underline{\lambda = 0}$ The boundary conditions produce

$$\phi(0) = c_1 = 0$$
$$\frac{d\phi}{dx}(L) = c_2 = 0$$

Leaving only the trivial solution, so $\lambda = 0$ is not an eigenvalue.

$\underline{\lambda < 0}$ Let $s = -\lambda$. The boundary conditions produce

$$\phi(0) = c_1 = 0$$
$$\frac{d\phi}{dx}(L) = c_2\sqrt{s}\cosh\left(\sqrt{s}L\right) = 0 \qquad \Longrightarrow c_2 = 0$$

which confirms that $\lambda < 0$ are ineligible to be eigenvalues of the boundary value problem.

(e) $\underline{\lambda > 0}$ The boundary conditions produce

$$\frac{d\phi}{dx}(0) = c_2\sqrt{\lambda} = 0 \qquad \Longrightarrow c_2 = 0$$
$$\phi(L) = c_1\cos\left(\sqrt{\lambda}L\right) = 0$$

which forces $\lambda = [(n-1/2)\,\pi/L]^2$ for positive integers. Then

$$\phi(x) = c_1 \cos\left(\frac{\left(n-\frac{1}{2}\right)\pi x}{L}\right)$$

$\underline{\lambda = 0}$ The boundary conditions produce

$$\frac{d\phi}{dx}(0) = c_2 = 0$$
$$\phi(L) = c_1 = 0$$

implying that $\lambda = 0$ cannot be an eigenvalue of ϕ.

$\underline{\lambda < 0}$ Let $s = -\lambda$. The boundary conditions produce

$$\frac{d\phi}{dx}(0) = c_2\sqrt{s} = 0 \qquad \Longrightarrow c_2 = 0$$
$$\phi(L) = c_1\cosh\left(\sqrt{s}L\right) = 0 \qquad \Longrightarrow c_1 = 0$$

once more indicating that the trivial solution is all that is yielded, ensuring that $\lambda < 0$ can never be eigenvalues of the boundary value problem.

(f) $\underline{\lambda > 0}$ The boundary conditions produce

$$\phi(a) = c_1 \cos\left(\sqrt{\lambda}a\right) + c_2 \sin\left(\sqrt{\lambda}a\right) = 0$$
$$\phi(b) = c_1 \cos\left(\sqrt{\lambda}b\right) + c_2 \sin\left(\sqrt{\lambda}b\right) = 0$$

One insight here is to rewrite these equations as a matrix equation

$$\begin{bmatrix} \cos\left(\sqrt{\lambda}a\right) & \sin\left(\sqrt{\lambda}a\right) \\ \cos\left(\sqrt{\lambda}b\right) & \sin\left(\sqrt{\lambda}b\right) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

And calculate the determinant of the left-most matrix using the sine angle difference identity:

$$\det = \cos\left(\sqrt{\lambda}a\right) \sin\left(\sqrt{\lambda}b\right) - \cos\left(\sqrt{\lambda}b\right) \sin\left(\sqrt{\lambda}a\right) = \sin\left(\sqrt{\lambda}(b-a)\right)$$

Now, the question before us that will determine λ: is the determinant zero or non-zero? We know that we cannot have c_1 and c_2 both equal to zero if we want positive eigenvalues, which implies that there exists a non-trivial solution. In turn, we can deduce that *the rows are linearly dependent*, guaranteeing a zero determinant. This forces

$$\lambda = \left(\frac{n\pi}{b-a}\right)^2$$

for positive integers. Since there exist non-zero coefficients c_1 and c_2, we can guess one solution for both

$$c_1 = -C \sin\left(\sqrt{\lambda}a\right)$$
$$c_2 = C \cos\left(\sqrt{\lambda}a\right)$$

In the first row the combination of the two terms is clearly zero, and in the second, that is simply equal to the determinant, which we have established is zero. The corresponding eigenfunctions are

$$\begin{aligned} \phi(x) &= -C \sin\left(\sqrt{\lambda}a\right) \cos\left(\sqrt{\lambda}x\right) + C \cos\left(\sqrt{\lambda}a\right) \sin\left(\sqrt{\lambda}x\right) \\ &= C \sin\left(\sqrt{\lambda}(x-a)\right) \\ &= C \sin\left(\frac{n\pi(x-a)}{b-a}\right) \end{aligned}$$

(g) $\underline{\lambda > 0}$ The boundary conditions produce

$$\phi(0) = c_1 = 0$$
$$\frac{d\phi}{dx}(L) + \phi(L) = c_2\sqrt{\lambda} \cos\left(\sqrt{\lambda}L\right) + c_2 \sin\left(\sqrt{\lambda}L\right) = 0$$

Equivalently, the second boundary condition can be written as

$$\tan\left(\sqrt{\lambda}L\right) = -\sqrt{\lambda}$$

which is legal as if $\cos\left(\sqrt{\lambda}L\right) = 0$, the corresponding values of λ would not render $\sin\left(\sqrt{\lambda}L\right) = 0$, forcing $c_2 = 0$, and producing only the trivial solution. The equation derived is a transcendental equation and must be evaluated with numerical methods. Referring to Figure 5.8.1 in Haberman demonstrates that there are infinitely many solutions. For now, call the solutions λ_1, λ_2, ..., where $\lambda_1 < \lambda_2 < \cdots$. Then $\phi_n(x) = c_2 \sin\left(\sqrt{\lambda_n}x\right)$ for positive integers.

$\underline{\lambda = 0}$ The boundary conditions produce

$$\phi(0) = c_1 = 0$$
$$\frac{d\phi}{dx}(L) + \phi(L) = c_2(1 + L) = 0$$

Forcing $c_2 = 0$, so only the trivial solution exists, implying $\lambda = 0$ cannot be an eigenvalue.

$\underline{\lambda < 0}$ Let $s = -\lambda$. The boundary conditions produce

$$\phi(0) = c_1 = 0$$
$$\frac{d\phi}{dx}(L) + \phi(L) = c_2\sqrt{s}\cosh\left(\sqrt{s}L\right) + c_2\sinh\left(\sqrt{s}L\right) = 0$$

Again, we can rewrite the second boundary condition as

$$\tanh\left(\sqrt{s}L\right) = -\sqrt{s}$$

This is once more a transcendental equation, however it is true only at $s = 0$, as when $s > 0$, $\tanh(\sqrt{s}) > 0$ and $-\sqrt{s} < 0$. Therefore we have $c_2 = 0$, and once again only the trivial solution. There exist no eigenvalues $\lambda < 0$ for ϕ.

Question: 2.3.3

Consider the heat equation

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$$

subject to the boundary conditions

$$u(0,t) = 0 \qquad \text{and} \qquad u(L,t) = 0$$

Solve the initial value problem if the temperature is initially

(a) $u(x,0) = 6\sin\left(\frac{9\pi x}{L}\right)$

(b) $u(x,0) = 3\sin\left(\frac{\pi x}{L}\right) - \sin\left(\frac{3\pi x}{L}\right)$

(c) $u(x,0) = 2\cos\left(\frac{3\pi x}{L}\right)$

(d) $u(x,0) = \begin{cases} 1 & 0 < x \le L/2 \\ 2 & L/2 < x < L \end{cases}$

(e) $u(x,0) = f(x)$

Use the general solution

$$u(x,t) = B\sin\left(\frac{n\pi x}{L}\right)e^{-k(n\pi/L)^2 t}$$

(a) Set $B = 6$ and $n = 9$ to get

$$u(x,t) = 6\sin\left(\frac{9\pi x}{L}\right)e^{-k(9\pi/L)^2 t}$$

(b) From the principle of superposition, the general form looks like

$$u(x,t) = B_1\sin\left(\frac{n_1\pi x}{L}\right)e^{-k(n_1\pi/L)^2 t} + B_2\sin\left(\frac{n_2\pi x}{L}\right)e^{-k(n_2\pi/L)^2 t}$$

Set $B_1 = 3$, $B_2 = -1$, $n_1 = 1$, and $n_2 = 3$ to get

$$u(x,t) = 3\sin\left(\frac{\pi x}{L}\right)e^{-k(\pi/L)^2 t} - \sin\left(\frac{3\pi x}{L}\right)e^{-k(3\pi/L)^2 t}$$

(c) The issue here is the initial temperature fails to meet the boundary conditions. In this case, the general solution is

$$u(x,t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right)e^{-k(n\pi/L)^2 t}$$

From the initial condition, we must set equal

$$u(x,0) = 2\cos\left(\frac{3\pi x}{L}\right) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right)$$

Now, multiply both sides by $\sin(m\pi x/L)$ and integrate from 0 to L:

$$\int_0^L 2\cos\left(\frac{3\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)\,dx = \int_0^L \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)\,dx$$

By orthogonality of sines, the right-hand side equals

$$A_m \int_0^L \sin^2\left(\frac{m\pi x}{L}\right)\,dx$$

And solving for A_m gives us

$$A_m = \frac{\int_0^L 2\cos\left(\frac{3\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)\,dx}{\int_0^L \sin^2\left(\frac{m\pi x}{L}\right)\,dx} = \frac{2}{L}\int_0^L 2\cos\left(\frac{3\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)\,dx$$

Using the product identity for cosine and sine, we further derive

$$\begin{aligned}
A_m &= \frac{2}{L}\int_0^L 2\cos\left(\frac{3\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)\,dx \\
&= \frac{2}{L}\left[\int_0^L \sin\left(\frac{(m+3)\,\pi x}{L}\right)\,dx + \int_0^L \sin\left(\frac{(m-3)\,\pi x}{L}\right)\,dx\right] \\
&= -\frac{2}{L}\left[\frac{L}{(m+3)\,\pi}\cos\left(\frac{(m+3)\,\pi x}{L}\right)\Big|_0^L + \frac{L}{(m-3)\,\pi}\cos\left(\frac{(m-3)\,\pi x}{L}\right)\Big|_0^L\right] \\
&= \begin{cases} -\frac{2}{L}\left[-\frac{2L}{(m+3)\,\pi} - \frac{2L}{(m-3)\,\pi}\right], & m \text{ even} \\ 0, & m \text{ odd} \end{cases} \\
&= \begin{cases} \frac{8m}{(m+3)(m-3)\,\pi}, & m \text{ even} \\ 0, & m \text{ odd} \end{cases}
\end{aligned}$$

Rewriting as

$$A_n = \frac{16n}{(2n+3)(2n-3)\,\pi}$$

we have temperature distribution

$$u(x,t) = \sum_{n=1}^{\infty} \frac{16n}{(2n+3)(2n-3)\,\pi}\sin\left(\frac{2n\pi x}{L}\right)e^{-k(2n\pi/L)^2 t}$$

Observe that by ensuring that we sum only over even indices, *every* index throughout the expression must be scaled by a factor of two.

(d) From the general solution

$$u(x,t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right)e^{-k(n\pi/L)^2 t}$$

Set equal

$$u(x,0) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right)$$

And multiply both sides by $\sin(m\pi x/L)$ and integrate from 0 to L to get

$$\begin{aligned}
\int_0^L u(x,0)\sin\Big(\frac{m\pi x}{L}\Big)\,dx &= \int_0^{L/2}\sin\Big(\frac{m\pi x}{L}\Big)\,dx + \int_{L/2}^L 2\sin\Big(\frac{m\pi x}{L}\Big)\,dx \\
&= \int_0^L \sum_{n=1}^{\infty} A_n \sin\Big(\frac{n\pi x}{L}\Big)\sin\Big(\frac{m\pi x}{L}\Big)\,dx \\
&= A_m \int_0^L \sin^2\Big(\frac{m\pi x}{L}\Big)\,dx \\
&= A_m \frac{L}{2}
\end{aligned}$$

Isolating A_m, we find

$$\begin{aligned}
A_m &= \frac{2}{L}\left[\int_0^{L/2}\sin\Big(\frac{m\pi x}{L}\Big)\,dx + \int_{L/2}^L 2\sin\Big(\frac{m\pi x}{L}\Big)\,dx\right] \\
&= -\frac{2}{L}\frac{L}{m\pi}\left[\cos\Big(\frac{m\pi x}{L}\Big)\Big|_0^{L/2} + 2\cos\Big(\frac{m\pi x}{L}\Big)\Big|_{L/2}^{L}\right] \\
&= \frac{2}{m\pi}\left[1+\cos\Big(\frac{m\pi}{2}\Big) - 2\cos(m\pi)\right]
\end{aligned}$$

Unfortunately, due to the $\cos(m\pi/2)$ term and the modulo 4 arithmetic involved in evaluating it, there is no simpler algebraic expression for this Fourier coefficient. Nevertheless, it does allow us to write the complete temperature distribution:

$$u(x,t) = \sum_{n=1}^{\infty} \frac{2}{n\pi}\left[1+\cos\Big(\frac{n\pi}{2}\Big) - 2\cos(n\pi)\right]\sin\Big(\frac{n\pi x}{L}\Big)\, e^{-k(n\pi/L)^2 t}$$

(e) In the absolute general case for the initial condition, we set equal

$$u(x,0) = f(x) = \sum_{n=1}^{\infty} A_n \sin\Big(\frac{n\pi x}{L}\Big)$$

Multiply both sides by $\sin(m\pi x/L)$, integrate over 0 to L, and invoke sine orthongonality to get

$$\begin{aligned}
\int_0^L f(x)\sin\Big(\frac{m\pi x}{L}\Big)\,dx &= \int_0^L \sum_{n=1}^{\infty} A_n \sin\Big(\frac{n\pi x}{L}\Big)\sin\Big(\frac{m\pi x}{L}\Big)\,dx \\
&= A_m \int_0^L \sin^2\Big(\frac{m\pi x}{L}\Big)\,dx
\end{aligned}$$

Leading us to

$$A_m = \frac{\displaystyle\int_0^L f(x)\sin\Big(\frac{m\pi x}{L}\Big)\,dx}{\displaystyle\int_0^L \sin^2\Big(\frac{m\pi x}{L}\Big)\,dx} = \frac{2}{L}\int_0^L f(x)\sin\Big(\frac{m\pi x}{L}\Big)\,dx$$

Question: 2.3.4

Consider

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$$

subject to $u(0,t) = 0$, $u(L,t) = 0$, and $u(x,0) = f(x)$.

(a) What is the total heat energy in the rod as a function of time?

(b) What is the flow of heat energy out of the rod at $x = 0$? at $x = L$?

(c) What relationship should exist between parts (a) and (b)?

(a) For the one-dimensional rod, we have the energy per unit length, $c\rho Au(x,t)$. To find the total energy in the rod, we must integrate across the entire length 0 to L. Let B_n be the Fourier coefficient derived in 2.3.3(e). Compute

$$\begin{aligned}
\text{Total Energy} &= \int_0^L c\rho Au(x,t)\ dx \\
&= \int_0^L c\rho A \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t}\, dx \\
&= c\rho A \sum_{n=1}^{\infty} B_n \frac{L}{n\pi}[1 - \cos(n\pi)]\, e^{-k(n\pi/L)^2 t} \\
&= c\rho A \sum_{j=1}^{\infty} B_{2j-1} \frac{2L}{(2j-1)\,\pi} e^{-k((2j-1)\pi/L)^2 t}
\end{aligned}$$

Where in the final step, we re-index the sum based on whether n is even or odd.

(b) From the Fourier heat transfer law, we multiply by the area A to get the total flow of energy per time

$$\phi(x)\,A = -k_0\frac{\partial u}{\partial x} = -k_0 A \sum_{n=1}^{\infty} B_n \frac{n\pi}{L} \cos\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t}$$

At either end

$$\begin{aligned}
\phi(0)\,A &= -k_0 A \sum_{n=1}^{\infty} B_n \frac{n\pi}{L} e^{-k(n\pi/L)^2 t} \\
\phi(L)\,A &= -k_0 A \sum_{n=1}^{\infty} B_n \frac{n\pi}{L} (-1)^n\, e^{-k(n\pi/L)^2 t}
\end{aligned}$$

where the sign convention is $\phi(x)$ describes the *rightward* flow of heat energy per time per unit area at point x.

(c) Physically, we can reconstruct the heat equation by balancing the rate of change of total energy with the spatial difference in flux at both ends of the rod. For the former, if E is the total energy and $k = k_0/c\rho$, then

$$\begin{aligned}\frac{\partial E}{\partial t} &= -c\rho k A \sum_{j=1}^{\infty} B_{2j-1} \left(\frac{(2j-1)\pi}{L}\right)^2 \left(\frac{2L}{(2j-1)\pi}\right) e^{-k((2j-1)\pi/L)^2 t} \\ &= -k_0 A \sum_{j=1}^{\infty} 2B_{2j-1} \frac{(2j-1)\pi}{L} e^{-k((2j-1)\pi/L)^2 t}\end{aligned}$$

and for the latter

$$\begin{aligned}[\phi(0) - \phi(L)]\, A &= -k_0 A \sum_{n=1}^{\infty} B_n \frac{n\pi}{L} e^{-k(n\pi/L)^2 t} (1 - (-1)^n) \\ &= -k_0 A \sum_{j=1}^{\infty} 2B_{2j-1} \frac{(2j-1)\pi}{L} e^{-k((2j-1)\pi/L)^2 t}\end{aligned}$$

where the last line arises from the elimination of the even modes, leaving only the odd modes. Hence the rate of change of energy balances the spatial difference of heat flux:

$$\frac{\partial E}{\partial t} = [\phi(0) - \phi(L)]\, A$$

Question: 2.3.5

Evaluate (be careful if $n = m$)

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx \qquad \text{for } n > 0, m > 0$$

Use the trigonometric identity

$$\sin(a)\sin(b) = \frac{1}{2}[\cos(a-b) - \cos(a+b)]$$

From the identity, if $n \neq m$, solve

$$\begin{aligned}\int_0^L \frac{1}{2}\left[\cos\left(\frac{(n-m)\pi x}{L}\right) - \cos\left(\frac{(n+m)\pi x}{L}\right)\right] dx &= \frac{1}{2}\left[\frac{L}{(n-m)\pi} \sin\left(\frac{(n-m)\pi x}{L}\right) - \right. \\ &\quad \left. \frac{L}{(n+m)\pi} \sin\left(\frac{(n+m)\pi x}{L}\right)\right]\Big|_0^L \\ &= 0\end{aligned}$$

which proves the orthogonality of sines when $n \neq m$. In the case when $n = m$, solve using the double angle identity:

$$\begin{aligned}\int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx &= \int_0^L \frac{1}{2}\left(1 - \cos\left(\frac{2n\pi x}{L}\right)\right) dx \\ &= \frac{x}{2}\Big|_0^L - \frac{L}{4n\pi} \cos\left(\frac{2n\pi x}{L}\right)\Big|_0^L \\ &= \frac{L}{2}\end{aligned}$$

Explaining the $L/2$ factor that arises when calculating Fourier coefficients.

Question: 2.3.6

Evaluate

$$\int_0^L \cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)\,dx \qquad \text{for } n \geq 0, m \geq 0$$

Use the trigonometric identity

$$\cos(a)\cos(b) = \frac{1}{2}[\cos(a+b) + \cos(a-b)]$$

(Be careful if $a - b = 0$ or $a + b = 0$.)

Suppose we pick n and m such that $n - m \neq 0$ and $n + m \neq 0$. Then solve

$$\begin{aligned}\int_0^L \frac{1}{2}\left[\cos\left(\frac{(n+m)\,\pi x}{L}\right) + \cos\left(\frac{(n-m)\,\pi x}{L}\right)\right]\,dx &= \frac{1}{2}\Bigg[\frac{L}{(n+m)\,\pi}\sin\left(\frac{(n+m)\,\pi x}{L}\right) + \\ &\qquad \frac{L}{(n-m)\,\pi}\sin\left(\frac{(n-m)\,\pi x}{L}\right)\Bigg]\Bigg|_0^L \\ &= 0\end{aligned}$$

which demonstrates the orthogonality of cosines. In the case when $m = n \neq 0$, then

$$\begin{aligned}\int_0^L \cos^2\left(\frac{n\pi x}{L}\right)\,dx &= \int_0^L \frac{1}{2}\left[1 + \cos\left(\frac{2n\pi x}{L}\right)\right]\,dx \\ &= \frac{x}{2}\Big|_0^L + \frac{L}{2n\pi}\sin\left(\frac{2n\pi x}{L}\right)\Big|_0^L \\ &= \frac{L}{2}\end{aligned}$$

Again demonstrating that the squared norm of each nonzero cosine mode on $[0, L]$ is $L/2$. Now, in the special case when $m = n = 0$, both cosines are 1, and

$$\int_0^L dx = L$$

We are not yet at this point in the textbook, but it is this result that explains the $a_0 = \frac{1}{L}\int_0^L f(x)\,dx$ term in the Fourier cosine series.

Question: 2.3.7

Consider the following boundary value problem (if necessary, see Section 2.4.1):

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2} \text{ with } \frac{\partial u}{\partial x}(0,t) = 0, \frac{\partial u}{\partial x}(L,t) = 0, \text{ and } u(x,0) = f(x)$$

(a) Give a one-sentence physical interpretation of this problem.

(b) Solve by the method of separation of variables. First show that there are no separated solutions which exponentially grow in time. [*Hint*: The answer is

$$u(x,t) = A_0 + \sum_{n=1}^{\infty} A_n e^{-\lambda_n kt} \cos\left(\frac{n\pi x}{L}\right)]$$

What is λ_n?

(c) Show that the initial condition, $u(x,0) = f(x)$, is satisfied if

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)$$

(d) Using Exercise 2.3.6, solve for A_0 and A_n $(n \geq 1)$.

(e) What happens to the temperature distribution as $t \to \infty$? Show that it approaches the steady-state temperature distribution (see Section 1.4).

(a) The ends of the rod are insulated.

(b) Let $u(x,t) = \phi(x)\,G(t)$. Then

$$\frac{\partial u}{\partial t} = \phi(x)\,\frac{dG}{dt}$$
$$k\frac{\partial^2 u}{\partial x^2} = k\frac{d^2\phi}{dx^2}G(t)$$

Separating variables, derive

$$\frac{1}{kG(t)}\frac{dG}{dt} = \frac{1}{\phi(x)}\frac{d^2\phi}{dx^2} = -\lambda$$

From the time equation, we get

$$G(t) = ce^{-\lambda kt}$$

and from the spatial equation, we must solve by assuming different values of λ. First, consider $\lambda < 0$, and let $s = -\lambda$. Then $s > 0$. We solve

$$\frac{d^2\phi}{dx^2} = -\lambda\phi(x)$$

and find
$$\phi(x) = c_1 \cosh\left(\sqrt{s}x\right) + c_2 \sinh\left(\sqrt{s}x\right)$$
Now, we must verify that we can find non-trivial solutions subject to the boundary conditions $\partial\phi/\partial x(0,t) = 0$ and $\partial\phi/\partial x(L,t) = 0$. First, the spatial derivative
$$\frac{\partial\phi}{\partial x} = c_1\sqrt{s}\sinh\left(\sqrt{s}x\right) + c_2\sqrt{s}\cosh\left(\sqrt{s}x\right)$$
And evaluating the boundary conditions, we find
$$\frac{\partial\phi}{\partial x}(0) = c_2\sqrt{s} = 0$$
which implies $c_2 = 0$. For the second,
$$\frac{\partial\phi}{\partial x}(L) = c_1\sqrt{s}\sinh\left(\sqrt{s}L\right) = 0$$
which must imply $c_1 = 0$; for $\sinh(\sqrt{s}L) = 0$, we would require $\sqrt{s}L = 0$, but that cannot be the case. Therefore, only the trivial solution remains. Now, $G(t) = ce^{-\lambda kt}$ grows only when $\lambda < 0$, but since $\lambda < 0$ only admits the trivial solution, we have now established that there cannot be any non-trivial solution that exponentially grows in time.

In the case where $\lambda = 0$, we solve the ordinary differential equation to get
$$\phi(x) = c_1 + c_2 x$$
and imposing the boundary conditions produces
$$\frac{\partial\phi}{\partial x}(0) = \frac{\partial\phi}{\partial x}(L) = c_2 = 0$$
which implies constant $\phi(x)$ and $G(t)$ and thus, constant temperature throughout the rod. We will include this constant term in our solution shortly.

Lastly, when $\lambda > 0$, we have
$$\phi(x) = c_1\cos\left(\sqrt{\lambda}x\right) + c_2\sin\left(\sqrt{\lambda}x\right)$$
which has spatial derivative
$$\frac{\partial\phi}{\partial x} = -c_1\sqrt{\lambda}\sin\left(\sqrt{\lambda}x\right) + c_2\sqrt{\lambda}\cos\left(\sqrt{\lambda}x\right)$$
Imposing the boundary conditions yield
$$\frac{\partial\phi}{\partial x}(0) = c_2\sqrt{\lambda} = 0 \implies c_2 = 0, \qquad \frac{\partial\phi}{\partial x}(L) = -c_1\sqrt{\lambda}\sin\left(\sqrt{\lambda}L\right) = 0$$
where in the second boundary condition, if we hold $c_1 \neq 0$, a non-trivial solution emerges if we set
$$\lambda_n = \left(\frac{n\pi}{L}\right)^2$$

where n is any positive integer. Since the solution holds for any cosine mode n, appealing to superposition, we can write our complete solution by combining the separated functions into a linear combination:

$$u(x,t) = A_0 + \sum_{n=1}^{\infty} A_n e^{-\lambda_n kt} \cos\left(\frac{n\pi x}{L}\right)$$

where A_0 is our constant solution from the zero eigenvalue case, λ_n our indexed eigenvalue, and A_n arbitrary constants determined by the initial condition $f(x)$.

(c) From part (b), at $t = 0$, we have

$$u(x,0) = f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)$$

(d) For A_0, since L is the squared norm of cosine at $n = m = 0$, we simply write

$$A_0 = \frac{\int_0^L f(x)\cos(0)\ dx}{\int_0^L\ dx} = \frac{1}{L}\int_0^L f(x)\ dx$$

As for the remaining coefficients, we have

$$A_n = \frac{\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx}{\int_0^L \cos^2\left(\frac{n\pi x}{L}\right)\ dx} = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

(e) In the limit, we can evaluate

$$\lim_{t\to\infty} u(x,t) = \lim_{t\to\infty}\left[A_0 + \sum_{n=1}^{\infty} A_n e^{-\lambda_n kt}\cos\left(\frac{n\pi x}{L}\right)\right] = A_0$$

as demonstrated in equation (1.4.21), the steady-state distribution in the case of insulated ends produces the average of the initial temperature distribution:

$$A_0 = \frac{1}{L}\int_0^L f(x)\ dx$$

Question: 2.3.8

Consider

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2} - \alpha u$$

This corresponds to a one-dimensional rod either with heat loss through the lateral sides with outside temperature $0°$ ($\alpha > 0$, see Exercise 1.2.4) or with insulated lateral sides with a heat sink proportional to the temperature. Suppose that the boundary conditions are

$$u(0,t) = 0 \text{ and } u(L,t) = 0$$

(a) What are the possible equilibrium temperature distributions if $\alpha > 0$?

(b) Solve the time-dependent problem $[u(x,0) = f(x)]$ if $\alpha > 0$. Analyze the temperature for large time ($t \to \infty$) and compare part (a).

(a) Let $\partial u/\partial t = 0$. Then isolating the second derivative, we solve

$$\frac{d^2 u}{dx^2} = \frac{\alpha}{k} u$$

Let $u = e^{rx}$. Then the characteristic polynomial $r^2 - \alpha/k = 0$ emerges, producing roots $r = \pm\sqrt{\alpha/k}$. The general solution is

$$u(x) = c_1 e^{\sqrt{\alpha/k}x} + c_2 e^{-\sqrt{\alpha/k}x}$$

The boundary conditions produce the equations

$$\begin{aligned} u(0) &= c_1 + c_2 = 0 \\ u(L) &= c_1 e^{\sqrt{\alpha/k}L} + c_2 e^{-\sqrt{\alpha/k}L} = 0 \end{aligned}$$

If you write out the matrix equation, it is easy to see that the two columns of the coefficient matrix are linearly independent; the first entries are both 1 while the second entries differ. Therefore only the trivial solution exists. More concretely, from the first equation we deduce $c_1 = -c_2$. Substitute into the second equation to get

$$u(L) = c_1\left[e^{\sqrt{\alpha/k}L} - e^{-\sqrt{\alpha/k}L}\right] = 0$$

The bracketed term is zero only when $\alpha = 0$, so that forces $c_1 = 0$. Immediately from the first equation, it follows that $c_2 = 0$ as well. Therefore $u(x) = 0$ is the only permissible temperature distribution at equilibrium.

(b) Use the method of separation of variables. Let $u(x,t) = \phi(x)\,G(t)$. Then after some substitution and division, derive

$$\frac{1}{kG(t)}\frac{dG}{dt} = \frac{1}{\phi(x)}\frac{d^2\phi}{dx^2} - \frac{\alpha}{k} = -\lambda$$

The time equation is derived similarly as before:

$$G(t) = ce^{-\lambda k t}$$

The spatial equation, on the other hand, looks like

$$\frac{d^2\phi}{dx^2} = \left[-\lambda + \frac{\alpha}{k}\right]\phi(x)$$

There is no difference in technique, we need only handle the additional α/k term. To do so, consider first the case where $\lambda < \alpha/k$. Then

$$\phi(x) = c_1 \cosh\left(\left(\sqrt{\frac{\alpha}{k} - \lambda}\right) x\right) + c_2 \sinh\left(\left(\sqrt{\frac{\alpha}{k} - \lambda}\right) x\right)$$

Imposing the boundary conditions produces

$$\phi(0) = c_1 = 0, \qquad \phi(L) = c_2 \sinh\left(\left(\sqrt{\frac{\alpha}{k} - \lambda}\right) L\right) = 0$$

Since the argument of the sinh term is positive, it follows that the term itself cannot be zero, forcing $c_2 = 0$. Therefore only the trivial solution arises.

Next, consider $\lambda = \alpha/k$. Then the right-hand side of the ordinary differential equation vanishes, and integrating twice for the general solution leads to

$$\phi(x) = c_1 + c_2 x$$

The boundary conditions once again enable us to conclude that only the trivial solution is viable:

$$\phi(0) = c_1 = 0, \qquad \phi(L) = c_2 L = 0 \implies c_2 = 0$$

Lastly, examine $\lambda > \alpha/k$. Then we have general solution

$$\phi(x) = c_1 \cos\left(\left(\sqrt{\lambda - \frac{\alpha}{k}}\right) x\right) + c_2 \sin\left(\left(\sqrt{\lambda - \frac{\alpha}{k}}\right) x\right)$$

Applying the boundary conditions yield

$$\phi(0) = c_1 = 0, \qquad \phi(L) = c_2 \sin\left(\left(\sqrt{\lambda - \frac{\alpha}{k}}\right) L\right) = 0$$

where in the second boundary condition, a non-trivial solution emerges by holding $c_2 \neq 0$ and setting

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2 + \frac{\alpha}{k}$$

for positive integers n. This yields eigenfunctions

$$\phi(x) = \sin\left(\frac{n\pi x}{L}\right)$$

By superposition and combining the spatial and temporal solutions, we derive the time-dependent temperature distribution

$$\begin{aligned} u(x,t) &= \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-\left[(n\pi/L)^2+\alpha/k\right]kt} \\ &= e^{-\alpha t} \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t} \end{aligned}$$

where B_n is defined as

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\, dx$$

In the temporal limit, we can see that

$$\lim_{t\to\infty} u(x,t) = \lim_{t\to\infty} e^{-\alpha t} \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t} = 0$$

which confirms our equilibrium result.

Question: 2.3.9

Redo Exercise 2.3.8 if $\alpha < 0$. [Be especially careful if $-\alpha/k = (n\pi/L)^2$.]

(a) When $\alpha < 0$, the roots of the characteristic polynomial are $r = \pm i\sqrt{-\alpha/k}$. The general solution now changes to

$$u(x) = c_1 \cos\left(\sqrt{-\frac{\alpha}{k}}x\right) + c_2 \sin\left(\sqrt{-\frac{\alpha}{k}}x\right)$$

The boundary conditions produce the equations

$$\begin{aligned} u(0) &= c_1 = 0 \\ u(L) &= c_2 \sin\left(\sqrt{-\frac{\alpha}{k}}L\right) = 0 \end{aligned}$$

where in the second equation, it is not the general case that the sine term is zero, so we must also have $c_2 = 0$. Thus in the general case, $u(x) = 0$, once again, is the only permissible temperature distribution at equilibrium.

However, consider the case when $-\alpha/k = (n\pi/L)^2$, where n is a positive integer. Then there exists a non-trivial solution, as the sine term goes to zero, and c_2 is left as a free parameter (contingent on the initial temperature). We are left with $u(x) = c_2 \sin(n\pi x/L)$.

(b) The derivation of the spatial equations proceeds exactly as in the case of positive α. The eigenvalues are also

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2 + \frac{\alpha}{k}$$

However, since $\alpha/k < 0$, we reach an interesting conclusion: $\lambda_n > 0$ is no longer a guarantee.

Let us take a step back. What does $\alpha < 0$ physically represent? Previously, we interpreted the $-\alpha u$ term as a heat sink. Reversing the coefficient sign turns this into a heat *source*. We have three possibilities:

- Net outward diffusion of heat energy exceeds internal generation of heat from the source, contributing to energy loss and decaying temperature over time.
- Net outward diffusion of heat energy perfectly balances internal heat generation, contributing to constant energy and constant temperature distribution over time.
- Net outward diffusion of heat energy is less than internal heat generation, contributing to energy growth and increasing temperature over time.

Consider the bracketed term $(n\pi/L)^2 + \alpha/k$ in the exponential:

$$u(x,t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k\left[(n\pi/L)^2 + \alpha/k\right]t}$$

A subtle point: we assume that α/k is some fixed quantity. Therefore, its ordering relative to $(n\pi/L)^2$ will depend on n. Conceptually, we must treat each mode on a case-by-case matter. For each respective possibility and each mode, we have

- Net outward diffusion of heat energy $>$ internal heat generation: $(n\pi/L)^2 > -\alpha/k$, $\lim_{t\to\infty} u_n(x,t) = 0$
- Net outward diffusion of heat energy $=$ internal heat generation: $(n\pi/L)^2 = -\alpha/k$, $\lim_{t\to\infty} u_n(x,t) = B_n \sin(n\pi x/L)$
- Net outward diffusion of heat energy $<$ internal heat generation: $(n\pi/L)^2 < -\alpha/k$, $\lim_{t\to\infty} u_n(x,t) = \infty$

where $u_n(x,t)$ corresponds to the n-th mode

$$u_n(x,t) = B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k\left[(n\pi/L)^2 + \alpha/k\right]t}$$

Observe that the limit of the case of equality is equivalent to the nontrivial solution derived in part (a): balancing of outward diffusion and internal generation leads to a steady-state temperature distribution.

Lastly, to find the limit of the *entire* temperature distribution $u(x,t)$, choose the smallest mode: $n = 1$. Since we fix α and k, at some point, $(n\pi/L)^2$ will exceed $-\alpha/k$, leading to temporal decay of those modes. What matters is *where* $(\pi/L)^2$ starts off *in relation* to $-\alpha/k$. If $(\pi/L)^2 > -\alpha/k$, then

$$\lim_{t\to\infty} u(x,t) = 0$$

Otherwise, in the cases when $(\pi/L)^2 = -\alpha/k$ and $(\pi/L)^2 < -\alpha/k$, we have, respectively

$$\lim_{t\to\infty} u(x,t) = B_1 \sin\left(\frac{\pi x}{L}\right) \qquad \lim_{t\to\infty} u(x,t) = \infty$$

Author's note: I love this problem. It is one of those problems where struggling will confer enormous insight. Who knew that reversing the sign on α would lend itself to such rich physics? My advice: reason through the conceptual physics first, before attempting to rush into mathematical manipulations. Thorough understanding of the physical system will provide the insight that very little manipulation is needed; subtle understanding of the behavior of each independent mode vis-à-vis the complete solution is essential.

Question: 2.3.10

For two- and three-dimensional vectors, the fundamental property of dot products, $\boldsymbol{A}\cdot\boldsymbol{B} = |\boldsymbol{A}||\boldsymbol{B}|\cos(\theta)$, implies that

$$|\boldsymbol{A}\cdot\boldsymbol{B}| \le |\boldsymbol{A}||\boldsymbol{B}| \qquad 2.3.44$$

In this exercise, we generalize this to n-dimensional vectors and functions, in which case (2.3.44) is known as **Schwarz's inequality**. [The names of Cauchy and Buniakovsky are also associated with (2.3.44).]

(a) Show that $|\boldsymbol{A} - \gamma\boldsymbol{B}|^2 \ge 0$ implies (2.3.44), where $\gamma = \boldsymbol{A}\cdot\boldsymbol{B}/\boldsymbol{B}\cdot\boldsymbol{B}$.

(b) Express the inequality using both

$$\boldsymbol{A}\cdot\boldsymbol{B} = \sum_{n=1}^{\infty} a_n b_n = \sum_{n=1}^{\infty} a_n c_n \frac{b_n}{c_n}$$

(c) Generalize (2.3.44) to functions. [*Hint*: Let $\boldsymbol{A}\cdot\boldsymbol{B}$ mean the integral $\int_0^L A(x)\,B(x)\,dx$.]

(a) Compute

$$\begin{aligned}
|\boldsymbol{A} - \gamma\boldsymbol{B}|^2 &= (\boldsymbol{A} - \gamma\boldsymbol{B})\cdot(\boldsymbol{A} - \gamma\boldsymbol{B}) \\
&= \boldsymbol{A}\cdot\boldsymbol{A} - 2\gamma\boldsymbol{A}\cdot\boldsymbol{B} + \gamma^2\boldsymbol{B}\cdot\boldsymbol{B} \\
&= |\boldsymbol{A}|^2 - \frac{2(\boldsymbol{A}\cdot\boldsymbol{B})^2}{|\boldsymbol{B}|^2} + \frac{(\boldsymbol{A}\cdot\boldsymbol{B})^2}{|\boldsymbol{B}|^2} \\
&\ge 0
\end{aligned}$$

Multiplying through by $|\boldsymbol{B}|^2$, collecting the $(\boldsymbol{A}\cdot\boldsymbol{B})^2$ terms and adding to the other side of the inequality, we get

$$|\boldsymbol{A}|^2|\boldsymbol{B}|^2 \ge (\boldsymbol{A}\cdot\boldsymbol{B})^2$$

taking the positive root of both sides, we conclude the Schwarz inequality:

$$|\boldsymbol{A}||\boldsymbol{B}| \ge |\boldsymbol{A}\cdot\boldsymbol{B}|$$

(b) From the squared vector definition of the Schwarz inequality, to express in algebraic form, we equivalently derive

$$\begin{aligned}(\boldsymbol{A}\cdot\boldsymbol{B})^2 &= \left(\sum_{n=1}^{\infty} a_n b_n\right)^2 \\ &\le \left(\sum_{n=1}^{\infty} a_n^2\right)\left(\sum_{n=1}^{\infty} b_n^2\right) \\ &= |\boldsymbol{A}|^2|\boldsymbol{B}|^2 \\ (\boldsymbol{A}\cdot\boldsymbol{B})^2 &= \left(\sum_{n=1}^{\infty} a_n c_n \frac{b_n}{c_n}\right)^2 \\ &\le \left(\sum_{n=1}^{\infty} a_n^2 c_n^2\right)\left(\sum_{n=1}^{\infty} \frac{b_n^2}{c_n^2}\right)\end{aligned}$$

The second inequality can be interpreted as a weighted inner product.

(c) Similarly, we proceed from the squared vector definition of the Schwarz inequality, and derive in integral form

$$\left(\int_0^L A(x)\,B(x)\,dx\right)^2 \le \int_0^L A(x)^2\,dx \int_0^L B(x)^2\,dx$$

Question: 2.3.11

Solve the heat equation $\dfrac{\partial u}{\partial t} = k\dfrac{\partial^2 u}{\partial x^2}$ subject to the following conditions:

$$u(0,t) = 0 \quad u(L,t) = 0 \quad u(x,0) = f(x)$$

What happens as $t \to \infty$? [*Hints*:

(1) It is known that if $u(x,t) = \phi(x)\,G(t)$, then $\dfrac{1}{kG}\dfrac{dG}{dt} = \dfrac{1}{\phi}\dfrac{\partial^2\phi}{\partial x^2}$.

(2) Use formula sheet.]

Let $u(x,t) = \phi(x)\,G(t)$. Then

$$\begin{aligned}\frac{\partial u}{\partial t} &= \phi(x)\,\frac{dG}{dt} \\ k\frac{\partial^2 u}{\partial x^2} &= k\frac{d^2\phi}{dx^2}G(t)\end{aligned}$$

Collecting similar variables and setting equal to a constant $-\lambda$, derive

$$\frac{1}{kG(t)}\frac{dG}{dt} = \frac{1}{\phi(x)}\frac{d^2\phi}{dx^2} = -\lambda$$

This produces time and spatial equations and we solve each individually. First the time equation:

$$G(t) = ce^{-\lambda k t}$$

And for the spatial equation, we consider three cases of λ: above, below, or at zero. Try $\lambda > 0$. Then $-\lambda < 0$, and after substituting $u = e^{rx}$, the characteristic polynomial yields complex roots. The ensuing linear combination of Euler equations can be rewritten as a linear combination of cosine and sine:

$$\phi(x) = c_1 \cos\left(\sqrt{\lambda}x\right) + c_2 \sin\left(\sqrt{\lambda}x\right)$$

Applying the boundary conditions, we find

$$\begin{aligned} \phi(0) &= c_1 = 0 \\ \phi(L) &= c_2 \sin\left(\sqrt{\lambda}L\right) = 0 \end{aligned}$$

where in the second boundary condition, if we assume $c_2 \neq 0$ so as not to be left with a trivial solution, and set $\lambda_n = (n\pi/L)^2$, then we have eigenvalues for the spatial function

$$\phi(x) = c_2 \sin\left(\frac{n\pi x}{L}\right)$$

Now, assume $\lambda = 0$. Then the spatial solution is

$$\phi(x) = c_1 + c_2 x$$

With boundary conditions $\phi(0) = c_1 = 0$ and $\phi(L) = c_2 L = 0$ implying $c_2 = 0$, we are left only with the trivial solution. Lastly, try $\lambda < 0$. Then set $-\lambda = s$, which is positive. With positive roots of the characteristic polynomial, we can express the spatial solution as a linear combination of hyperbolic cosine and sine:

$$\phi(x) = c_1 \cosh\left(\sqrt{s}x\right) + c_2 \sinh\left(\sqrt{s}x\right)$$

The boundary conditions produce $\phi(0) = c_1$ and $\phi(L) = c_2 \sinh(\sqrt{s}L) = 0$. Since $\sqrt{s}L \neq 0$, it is impossible for the hyperbolic sine to be zero, forcing $c_2 = 0$. Once more, the trivial solution is all that remains.

Therefore we are only permitted eigenvalues $\lambda_n > 0$.

Multiplying $\phi(x)$ and $G(t)$ and appealing to superposition, we have complete solution

$$u(x,t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t}$$

where the coefficients A_n are derived from the projection of the initial temperature $f(x)$ onto each of the sine modes spanning the Hilbert space of orthogonal sines:

$$A_n = \frac{\displaystyle\int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right)\, dx}{\displaystyle\int_0^L \sin^2\left(\frac{n\pi x}{L}\right)\, dx} = \frac{2}{L}\int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right)\, dx$$

Lastly, in the temporal limit, we have

$$\lim_{t\to\infty} u(x,t) = \lim_{t\to\infty} \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t} = 0$$

Physically, as we do not have insulation and the ends of the rod are assumed to be at zero temperature, the rod itself will eventually cool to the uniformly zero equilibrium temperature.

2.4 - Worked Examples with the Heat Equation (Other Boundary Value Problems)

Question: 2.4.1

Solve the heat equation $\partial u/\partial t = k\partial^2 u/\partial x^2$, $0 < x < L$, $t > 0$, subject to

$$\begin{aligned} \frac{\partial u}{\partial x}(0,t) &= 0 \qquad t > 0 \\ \frac{\partial u}{\partial x}(L,t) &= 0 \qquad t > 0 \end{aligned}$$

(a) $u(x,0) = \begin{cases} 0 & x < L/2 \\ 1 & x > L/2 \end{cases}$

(b) $u(x,0) = 6 + 4\cos\left(\frac{3\pi x}{L}\right)$

(c) $u(x,0) = -2\sin\left(\frac{\pi x}{L}\right)$

(d) $u(x,0) = -3\cos\left(\frac{8\pi x}{L}\right)$

For all solutions, use the general form

$$u(x,t) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) e^{-(n\pi/L)^2 kt}$$

where

$$u(x,0) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)$$

(a) Multiply the initial temperature by $\cos(m\pi x/L)$ and integrate from 0 to L to find

$$\begin{aligned} \int_0^L u(x,0)\cos\left(\frac{m\pi x}{L}\right) dx &= \int_0^{L/2} 0\cdot\cos\left(\frac{m\pi x}{L}\right) dx + \int_{L/2}^L 1\cdot\cos\left(\frac{m\pi x}{L}\right) dx \\ &= \int_0^L A_0\cos\left(\frac{m\pi x}{L}\right) dx + \int_0^L \sum_{n=1}^{\infty} A_n\cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right) dx \\ &= \begin{cases} A_0 L & m = 0 \\ A_m \displaystyle\int_0^L \cos^2\left(\frac{m\pi x}{L}\right) dx & m \geq 1 \end{cases} \end{aligned}$$

First isolate A_0 by setting $m = 0$, and derive

$$A_0 L = L - \frac{L}{2} = \frac{L}{2} \quad \Longrightarrow \quad A_0 = \frac{1}{2}$$

Next, isolating A_m, derive

$$\begin{aligned}
A_m &= 0 + \frac{\displaystyle\int_{L/2}^{L} \cos\left(\frac{m\pi x}{L}\right)\, dx}{\displaystyle\int_0^L \cos^2\left(\frac{m\pi x}{L}\right)\, dx} \\
&= -\frac{2}{m\pi}\sin\left(\frac{m\pi}{2}\right) \\
&= \begin{cases} 0 & m \text{ even} \\ -\dfrac{2}{m\pi}\sin\left(\dfrac{m\pi}{2}\right) & m \text{ odd} \end{cases}
\end{aligned}$$

Reindexing, the second case can be rewritten for all positive integers n as

$$A_n = (-1)^n \frac{2}{(2n-1)\,\pi}$$

Substituting into the general form, we have temperature distribution

$$u(x,t) = \frac{1}{2} + \sum_{n=1}^{\infty} (-1)^n \frac{2}{(2n-1)\,\pi} \cos\left(\frac{(2n-1)\,\pi x}{L}\right) e^{-((2n-1)\pi/L)^2 kt}$$

(b) The boundary conditions are satisfied, so we can conclude $A_0 = 6$, $n = 3$, and $A_3 = 4$.

(c) In this case, the boundary conditions are not satisfied. Multiply by $\cos(m\pi x/L)$ and integrate from 0 to L to derive

$$\begin{aligned}
\int_0^L u(x,0)\cos\left(\frac{m\pi x}{L}\right)\, dx &= -2\int_0^L \sin\left(\frac{\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)\, dx \\
&= \int_0^L A_0 \cos\left(\frac{m\pi x}{L}\right)\, dx + \int_0^L \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)\, dx \\
&= \begin{cases} A_0 L & m = 0 \\ A_m \displaystyle\int_0^L \cos^2\left(\frac{m\pi x}{L}\right)\, dx & m \geq 1 \end{cases}
\end{aligned}$$

First suppose $m = 0$. Then A_0 is

$$A_0 = -\frac{2}{L}\int_0^L \sin\left(\frac{\pi x}{L}\right)\, dx = -\frac{4}{\pi}$$

and for all other m, we have

$$\begin{aligned}
A_m &= -\frac{4}{L}\int_0^L \sin\left(\frac{\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)\,dx \\
&= -\frac{2}{L}\int_0^L \left[\sin\left(\frac{(1+m)\,\pi x}{L}\right) + \sin\left(\frac{(1-m)\,\pi x}{L}\right)\right]\,dx \\
&= \frac{2}{L}\left[\frac{L}{(1+m)\,\pi}\cos\left(\frac{(1+m)\,\pi x}{L}\right) + \frac{L}{(1-m)\,\pi}\cos\left(\frac{(1-m)\,\pi x}{L}\right)\right]\Big|_0^L \\
&= \frac{2}{\pi}\left[\frac{\cos((1+m)\,\pi)-1}{1+m} + \frac{\cos((1-m)\,\pi)-1}{1-m}\right] \\
&= \begin{cases} 0 & m \text{ odd} \\ -\dfrac{4}{\pi}\left[\dfrac{1}{1+m} + \dfrac{1}{1-m}\right] & m \text{ even} \end{cases}
\end{aligned}$$

Note that the $1-m$ term in the denominator shows up in the derivation, but since we eventually determined that the coefficient vanishes for all odd m, we need not worry about a division by zero. Reindexing the second case, we have for positive integers n

$$A_n = -\frac{8}{\pi}\left[\frac{1}{(1+2n)(1-2n)}\right] = \frac{8}{\pi(4n^2-1)}$$

Therefore, the complete temperature distribution is

$$u(x,t) = -\frac{4}{\pi} + \sum_{n=1}^{\infty}\frac{8}{\pi(4n^2-1)}\cos\left(\frac{2n\pi x}{L}\right)e^{-(2n\pi/L)^2kt}$$

(d) Lastly, the boundary conditions are met, so we can simply assign $A_0 = 0$, $n = 8$, and $A_8 = -3$.

Question: 2.4.2

Solve

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2} \quad \text{with} \quad \begin{aligned} &\frac{\partial u}{\partial x}(0,t) = 0 \\ &u(L,t) = 0 \\ &u(x,0) = f(x) \end{aligned}$$

For this problem you may assume that no solutions of the heat equation exponentially grow in time. You may also guess appropriate orthogonality conditions for the eigenfunctions.

In the case of mixed boundary conditions, proceed normally: let $u(x,t) = \phi(x)\,G(t)$. From the typical derivation, we have temporal solution $G(t) = ce^{-\lambda kt}$. For the spatial solutions, begin with $\lambda > 0$. Then the characteristic polynomial produces imaginary roots, with general solution

$$\phi(x) = c_1\cos\left(\sqrt{\lambda}x\right) + c_2\sin\left(\sqrt{\lambda}x\right)$$

Imposing the boundary conditions yield

$$\begin{aligned} \frac{d\phi}{dx}(0) &= c_2\sqrt{\lambda} = 0, \implies c_2 = 0 \\ \phi(L) &= c_1 \cos\left(\sqrt{\lambda}L\right) = 0 \end{aligned}$$

where we have a non-trivial solution if

$$\lambda_n = \left[\frac{(2n-1)\,\pi}{2L}\right]^2$$

Next, suppose $\lambda = 0$. Then

$$\phi(x) = c_1 + c_2 x$$

The boundary conditions give

$$\begin{aligned} \frac{d\phi}{dx}(0) &= c_2 = 0 \\ \phi(L) &= c_1 = 0 \end{aligned}$$

leaving us only with the trivial solution. Lastly, suppose $\lambda < 0$. Define $s = -\lambda$, so $s > 0$. The characteristic polynomial gives positive roots, so we have a linear combination of hyperbolic cosine and sine for the solution:

$$\phi(x) = c_1 \cosh\left(\sqrt{s}x\right) + c_2 \sinh\left(\sqrt{s}x\right)$$

and applying the boundary conditions

$$\begin{aligned} \frac{d\phi}{dx}(0) &= c_2\sqrt{s} = 0 \implies c_2 = 0 \\ \phi(L) &= c_1 \cosh\left(\sqrt{s}L\right) = 0 \end{aligned}$$

where the second boundary condition forces $c_1 = 0$, as the hyperbolic cosine is never zero. Thus we get the trivial solution once more. So we are left with the eigenfunctions

$$\phi_n(x) = c_1 \cos\left[\frac{(2n-1)\,\pi x}{2L}\right]$$

And lastly, appealing to superposition and combining with the temporal solution, we have temperature distribution

$$u(x,t) = \sum_{n=1}^{\infty} A_n \cos\left[\frac{(2n-1)\,\pi x}{2L}\right] e^{-[(2n-1)\pi/2L]^2 kt}$$

where the Fourier coefficients must be derived from projecting $f(x)$ onto the odd "half-modes" of cosine Hilbert space. We begin with

$$f(x) = \sum_{n=1}^{\infty} A_n \cos\left[\frac{(2n-1)\,\pi x}{2L}\right]$$

Now, multiply both sides by $\cos((2m-1)\,\pi x/2L)$ and integrate from 0 to L to get

$$\begin{aligned}
\int_0^L f(x)\cos\left[\frac{(2m-1)\,\pi x}{2L}\right]dx &= \int_0^L \sum_{n=1}^{\infty} A_n \cos\left[\frac{(2n-1)\,\pi x}{2L}\right]\cos\left[\frac{(2m-1)\,\pi x}{2L}\right]dx \\
&= A_m \int_0^L \cos^2\left[\frac{(2m-1)\,\pi x}{2L}\right]dx \\
&= A_m \frac{L}{2} \\
\implies A_m &= \frac{2}{L}\int_0^L f(x)\cos\left[\frac{(2m-1)\,\pi x}{2L}\right]dx
\end{aligned}$$

where the orthogonality condition still holds between differing values at the half-modes, enabling the truth of the second equality.

Question: 2.4.3

Solve the eigenvalue problem

$$\frac{d^2\phi}{dx^2} = -\lambda\phi$$

subject to

$$\phi(0) = \phi(2\pi) \quad \text{and} \quad \frac{d\phi}{dx}(0) = \frac{d\phi}{dx}(2\pi)$$

$\underline{\lambda > 0}$: We have general solution

$$\phi(x) = c_1\cos\left(\sqrt{\lambda}x\right) + c_2\sin\left(\sqrt{\lambda}x\right)$$

with boundary conditions

$$\begin{aligned}
\phi(0) = c_1 \quad &= c_1\cos\left(2\sqrt{\lambda}\pi\right) + c_2\sin\left(2\sqrt{\lambda}\pi\right) &&= \phi(2\pi) \\
\frac{d\phi}{dx}(0) = c_2\sqrt{\lambda} &= -c_1\sqrt{\lambda}\sin\left(2\sqrt{\lambda}\pi\right) + c_2\sqrt{\lambda}\cos\left(2\sqrt{\lambda}\pi\right) &&= \frac{d\phi}{dx}(2\pi)
\end{aligned}$$

This system of equations is solved if we observe that $\lambda_n = n^2$, for $n = 0,\ 1,\ \ldots$ is an eigenvalue for a corresponding non-trivial solution, namely

$$\phi_n(x) = c_1\cos(nx) + c_2\sin(nx)$$

More concretely, we can solve the matrix equation

$$\begin{bmatrix} 1-\cos\left(2\sqrt{\lambda}\pi\right) & -\sin\left(2\sqrt{\lambda}\pi\right) \\ \sqrt{\lambda}\sin\left(2\sqrt{\lambda}\pi\right) & \sqrt{\lambda}\left[1-\cos\left(2\sqrt{\lambda}\pi\right)\right] \end{bmatrix}\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

In order for the existence of a non-trivial solution, we must have a zero determinant, as that guarantees a non-trivial linear combination of the columns to yield the zero vector. Compute from the coefficient matrix

$$\begin{aligned}
\det &= \sqrt{\lambda}\Big[1-\cos\Big(2\sqrt{\lambda}\pi\Big)\Big]^2 + \sqrt{\lambda}\sin^2\Big(2\sqrt{\lambda}\pi\Big) \\
&= \sqrt{\lambda}\Big[1-2\cos\Big(2\sqrt{\lambda}\pi\Big) + \sin^2\Big(2\sqrt{\lambda}\pi\Big) + \cos^2\Big(2\sqrt{\lambda}\pi\Big)\Big] \\
&= \sqrt{\lambda}\Big[2-2\cos\Big(2\sqrt{\lambda}\pi\Big)\Big]
\end{aligned}$$

where the equality is zero if and only if $\lambda_n = n^2$ for $n = 0$, 1, ... Moreover, note that the entire coefficient matrix goes to rank zero, meaning that c_1 and c_2 are unconstrained. This leaves us with the aforementioned eigenfunctions ϕ_n.

$\underline{\lambda = 0}$: We have general solution

$$\phi(x) = c_1 + c_2 x$$

with boundary conditions

$$\begin{aligned}
\phi(0) = c_1 &= c_1 + 2c_2\pi = \phi(2\pi) \\
\frac{d\phi}{dx}(0) = c_2 &= c_2 \qquad\qquad = \frac{d\phi}{dx}(2\pi)
\end{aligned}$$

The boundary conditions force $c_2 = 0$, and c_1 is a free-standing constant. Thus for eigenvalue $\lambda_0 = 0$, we have corresponding eigenfunction

$$\phi_0 = c_1$$

$\underline{\lambda < 0}$: Let $s = -\lambda$ so $s > 0$. Then we have general solution

$$\phi(x) = c_1\cosh\big(\sqrt{s}x\big) + c_2\sinh\big(\sqrt{s}x\big)$$

with boundary conditions

$$\begin{aligned}
\phi(0) = c_1 \quad &= c_1\cosh\big(2\sqrt{s}\pi\big) + c_2\sinh\big(2\sqrt{s}\pi\big) \qquad = \phi(2\pi) \\
\frac{d\phi}{dx}(0) = c_2\sqrt{s} &= c_1\sqrt{s}\sinh\big(2\sqrt{s}\pi\big) + c_2\sqrt{s}\cosh\big(2\sqrt{s}\pi\big) = \frac{d\phi}{dx}(2\pi)
\end{aligned}$$

that correspond to the matrix equation

$$\begin{bmatrix} 1-\cosh(2\sqrt{s}\pi) & \sinh(2\sqrt{s}\pi) \\ -\sqrt{s}\sinh(2\sqrt{s}\pi) & \sqrt{s}[1-\cosh(2\sqrt{s}\pi)] \end{bmatrix}\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Computing the determinant of the coefficient matrix produces

$$\begin{aligned}
\det &= \sqrt{s}\big[1-\cosh\big(2\sqrt{s}\pi\big)\big]^2 + \sqrt{s}\sinh^2\big(2\sqrt{s}\pi\big) \\
&= \sqrt{s}\big[1-2\cosh\big(2\sqrt{s}\pi\big) + \sinh^2\big(2\sqrt{s}\pi\big) + \cosh^2\big(2\sqrt{s}\pi\big)\big] \\
&= \sqrt{s}\big[-2\cosh\big(2\sqrt{s}\pi\big) + 2\cosh^2\big(2\sqrt{s}\pi\big)\big] \\
&= 2\sqrt{s}\cosh\big(2\sqrt{s}\pi\big)\big[\cosh\big(2\sqrt{s}\pi\big) - 1\big]
\end{aligned}$$

which vanishes only when $2\sqrt{s}\pi = 0$, but since we specified positive s, we can only have the trivial solution: $c_1 = c_2 = 0$.

Question: 2.4.4

Explicitly show that there are no negative eigenvalues for

$$\frac{d^2\phi}{dx^2} = -\lambda\phi \quad \text{subject to} \quad \frac{d\phi}{dx}(0) = 0 \quad \text{and} \quad \frac{d\phi}{dx}(L) = 0$$

Let $\lambda < 0$. Then let $s = -\lambda$, so $s > 0$. We have general solution

$$\phi(x) = c_1 \cosh\left(\sqrt{s}x\right) + c_2 \sinh\left(\sqrt{s}x\right)$$

with boundary conditions

$$\frac{d\phi}{dx}(0) = c_2\sqrt{s} = 0 = c_1\sqrt{s}\sinh\left(\sqrt{s}L\right) + c_2\sqrt{s}\cosh\left(\sqrt{s}L\right) = \frac{d\phi}{dx}(L)$$

writing as the matrix equation

$$\begin{bmatrix} 0 & \sqrt{s} \\ \sqrt{s}\sinh(\sqrt{s}L) & \sqrt{s}\cosh(\sqrt{s}L) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the coefficient matrix, calculate the determinant, which if zero, guarantees a non-trivial pair (c_1, c_2):

$$\det = -s\sinh\left(\sqrt{s}L\right) = 0$$

which holds true only when $\sinh(\sqrt{s}L) = 0$, but this is impossible as that holds only at $\sqrt{s}L = 0$, and we assumed $s > 0$. Therefore no negative eigenvalues exist for this eigenfunction and boundary conditions.

Question: 2.4.5

This problem presents an alternative derivation of the heat equation for a thin wire. The equation for a circular wire of finite thickness is the two-dimensional heat equation (in polar coordinates). Show that this reduces to (2.4.25) if the temperature does not depend on r and if the wire is very thin.

The difficult part is in conceptualizing the problem. From section 1.5, recall that the polar heat equation can describe a disk or an annulus, depending on whether the lower bound for r is zero or not. In this scenario, imagine the annulus case, where the radial bounds a and b differ by some very small amount δ:

$$b - a = \delta << 1$$

It is this assumption that allows us to conclude that there is no radial variation in the temperature; put differently, r is constant. From (1.5.19), the two-dimensional heat equation in polar coordinates is

$$\frac{\partial u}{\partial t} = k\left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}\right)$$

Since temperature is independent of r, we immediately see that the first two terms vanish. Now we are left with

$$\frac{\partial u}{\partial t} = \frac{k}{r^2}\frac{\partial^2 u}{\partial \theta^2}$$

Recall that the arc length $s = r\theta$. The task will be to convert from (r, θ)-coordinates to s-coordinates. First compute

$$\frac{\partial r}{\partial s} = 0, \qquad \frac{\partial \theta}{\partial s} = \frac{1}{r}, \qquad \frac{\partial^2 \theta}{\partial s^2} = -\frac{1}{r^2}\frac{\partial r}{\partial s} = 0$$

where the first and last equalities arise from the result that r is constant. By the chain rule, derive (once again appealing to the premise of all radial derivatives vanishing)

$$\begin{aligned}
\frac{\partial u}{\partial s} &= \frac{\partial u}{\partial r}\frac{\partial r}{\partial s} + \frac{\partial u}{\partial \theta}\frac{\partial \theta}{\partial s} = \frac{\partial u}{\partial \theta}\frac{\partial \theta}{\partial s} \\
&= \left(\frac{1}{r}\frac{\partial}{\partial \theta}\right) u \\
\frac{\partial^2 u}{\partial s^2} &= \frac{\partial}{\partial s}\left(\frac{\partial u}{\partial \theta}\frac{\partial \theta}{\partial s}\right) \\
&= \frac{\partial^2 u}{\partial \theta^2}\left(\frac{\partial \theta}{\partial s}\right)^2 \\
&= \left(\frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}\right) u
\end{aligned}$$

Substituting back to the heat equation, make the final step to get

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial s^2}$$

Question: 2.4.6

Determine the equilibrium temperature distribution for the thin circular ring of Section 2.4.2:

(a) directly from the equilibrium problem (see Section 1.4)

(b) by comparing the limit as $t \to \infty$ of the time-dependent problem

(a) Set $\partial u/\partial t = 0$, leaving us with

$$\frac{\partial^2 u}{\partial x^2} = 0$$

Integrating twice gives us general solution

$$u(x) = c_1 + c_2 x$$

Imposing the boundary conditions produces

$$\begin{aligned}
u(-L) &= c_1 - c_2 L = c_1 + c_2 L = u(L) \\
\frac{du}{dx}(-L) &= c_2 \qquad = c_2 \qquad = \frac{du}{dx}(L)
\end{aligned}$$

which forces $c_2 = 0$, leaving only

$$u(x) = c_1$$

At equilibrium, the temperature in the ring is constant, with c_1 determined by the initial temperature (due to the conservation of thermal energy), namely

$$c_1 = \frac{1}{2L}\int_{-L}^{L} f(x)\, dx$$

(b) From equation (2.4.38)

$$u(x,t) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) e^{-(n\pi/L)^2 kt} + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) e^{-(n\pi/L)^2 kt}$$

Taking the temporal limit, it is simple to derive

$$\lim_{t\to\infty} u(x,t) = a_0$$

from which it follows that $a_0 = c_1$ from the previous part.

Question: 2.4.7

Solve the heat equation

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$$

[*Hints*: It is known that if $u(x,t) = \phi(x)\, G(t)$, then $\dfrac{1}{kG}\dfrac{dG}{dt} = \dfrac{1}{\phi}\dfrac{d^2\phi}{dx^2}$. Appropriately assume $\lambda > 0$. Assume the eigenfunctions $\phi_n(x)$ satisfy the following integral condition (orthogonality):

$$\int_0^L \phi_n(x)\, \phi_m(x)\, dx = \begin{cases} 0 & n \neq m \\ L/2 & n = m \end{cases}$$

subject to the following conditions:

(a) $u(0,t) = 0$, $u(L,t) = 0$, $u(x,0) = f(x)$

(b) $u(0,t) = 0$, $\dfrac{\partial u}{\partial x}(L,t) = 0$, $u(x,0) = f(x)$

(c) $\dfrac{\partial u}{\partial x}(0,t) = 0$, $u(L,t) = 0$, $u(x,0) = f(x)$

(d) $\dfrac{\partial u}{\partial x}(0,t) = 0$, $\dfrac{\partial u}{\partial x}(L,t) = 0$, $u(x,0) = f(x)$ and modify orthogonality condition [using Table 2.4.1.]

In all cases, the temporal solution is

$$G(t) = ce^{-\lambda kt}$$

and the spatial solution has general form

$$\phi(x) = c_1 \cos\left(\sqrt{\lambda}x\right) + c_2 \sin\left(\sqrt{\lambda}x\right)$$

(a) Impose the boundary conditions

$$\begin{aligned} \phi(0) &= c_1 &&= 0 \\ \phi(L) &= c_2 \sin\left(\sqrt{\lambda} L\right) &&= 0 \end{aligned}$$

where a non-trivial solution corresponds to eigenvalues

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad n = 1, 2, ...$$

Leaving us eigenfunctions

$$\phi_n(x) = \sin\left(\frac{n\pi x}{L}\right)$$

By superposition, we have temperature distribution

$$u(x,t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right) e^{-(n\pi/L)^2 kt}$$

where A_n is

$$A_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) \, dx$$

(b) Impose the boundary conditions

$$\begin{aligned} \phi(0) &= c_1 &&= 0 \\ \frac{d\phi}{dx}(L) &= c_2 \sqrt{\lambda} \cos\left(\sqrt{\lambda} L\right) &&= 0 \end{aligned}$$

where a non-trivial solution corresponds to

$$\lambda_n = \left[\frac{(2n-1)\,\pi}{2L}\right]^2, \quad n = 1, 2, ...$$

Leaving us eigenfunctions

$$\phi_n(x) = \sin\left[\frac{(2n-1)\,\pi x}{2L}\right]$$

By superposition, we have temperature distribution

$$u(x,t) = \sum_{n=1}^{\infty} A_n \sin\left[\frac{(2n-1)\,\pi x}{2L}\right] e^{-[(2n-1)\pi/2L]^2 kt}$$

where A_n is derived by projecting onto the Hilbert sine space of "half-modes" to get

$$\begin{aligned} \int_0^L f(x) \sin\left[\frac{(2n-1)\,\pi x}{2L}\right] \, dx &= \int_0^L \sum_{n=1}^{\infty} A_n \sin\left[\frac{(2n-1)\,\pi x}{2L}\right] \sin\left[\frac{(2m-1)\,\pi x}{2L}\right] \, dx \\ &= A_m \int_0^L \sin^2\left[\frac{(2m-1)\,\pi x}{2L}\right] \, dx \\ &= A_m \frac{L}{2} \\ \implies A_m &= \frac{2}{L} \int_0^L f(x) \sin\left[\frac{(2m-1)\,\pi x}{2L}\right] \, dx \end{aligned}$$

(c) Impose the boundary conditions

$$\begin{aligned} \frac{d\phi}{dx}(0) &= c_2\sqrt{\lambda} &&= 0 \implies c_2 = 0 \\ \phi(L) &= c_1 \cos\left(\sqrt{\lambda}L\right) = 0 \end{aligned}$$

where a non-trivial solution corresponds to eigenvalues

$$\lambda_n = \left[\frac{(2n-1)\,\pi}{2L}\right]^2, \quad n = 1, 2, ...$$

Leaving us eigenfunctions

$$\phi_n(x) = \cos\left[\frac{(2n-1)\,\pi x}{2L}\right]$$

By superposition, we have temperature distribution

$$u(x,t) = \sum_{n=1}^{\infty} A_n \cos\left[\frac{(2n-1)\,\pi x}{2L}\right] e^{-[(2n-1)\pi/2L]^2 kt}$$

where A_n is derived by projecting onto the Hilbert cosine space of "half-modes" to get

$$\begin{aligned} \int_0^L f(x)\cos\left[\frac{(2n-1)\,\pi x}{2L}\right]\,dx &= \int_0^L \sum_{n=1}^{\infty} A_n \cos\left[\frac{(2n-1)\,\pi x}{2L}\right]\cos\left[\frac{(2m-1)\,\pi x}{2L}\right]\,dx \\ &= A_m \int_0^L \cos^2\left(\frac{(2m-1)\,\pi x}{2L}\right)\,dx \\ &= A_m \frac{L}{2} \\ \implies A_m &= \frac{2}{L}\int_0^L f(x)\cos\left[\frac{(2m-1)\,\pi x}{2L}\right]\,dx \end{aligned}$$

(d) Impose the boundary conditions

$$\begin{aligned} \frac{d\phi}{dx}(0) &= c_2\sqrt{\lambda} &&= 0 \implies c_2 = 0 \\ \frac{d\phi}{dx}(L) &= -c_1\sqrt{\lambda}\sin\left(\sqrt{\lambda}L\right) = 0 \end{aligned}$$

where a non-trivial solution corresponds to eigenvalues

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad n = 0, 1, ...$$

Leaving us eigenfunctions

$$\phi_n(x) = \cos\left(\frac{n\pi x}{L}\right)$$

By superposition, we have temperature distribution

$$u(x,t) = \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) e^{-(n\pi/L)^2 kt}$$

where the orthogonality conditions for A_0 and A_n, $n \neq 0$, slightly differ:

$$A_n = \begin{cases} \dfrac{1}{L}\displaystyle\int_0^L f(x)\ dx & n = 0 \\ \dfrac{2}{L}\displaystyle\int_0^L f(x)\cos\Big(\dfrac{n\pi x}{L}\Big)\ dx & n \neq 0 \end{cases}$$

2.5 - Laplace's Equation: Solutions and Qualitative Properties

Question: 2.5.1

Solve Laplace's equation inside a rectangle $0 \leq x \leq L$, $0 \leq y \leq H$, with the following boundary conditions [*Hint*: Separate variables. If there are two homogeneous boundary conditions in y, let $u(x,y) = h(x)\,\phi(y)$, and if there are two homogeneous boundary conditions in x, let $u(x,y) = \phi(x)\,h(y)$.]:

(a) $\frac{\partial u}{\partial x}(0,y) = 0$, $\frac{\partial u}{\partial x}(L,y) = 0$, $u(x,0) = 0$, $u(x,H) = f(x)$

(b) $\frac{\partial u}{\partial x}(0,y) = g(y)$, $\frac{\partial u}{\partial x}(L,y) = 0$, $u(x,0) = 0$, $u(x,H) = 0$

(c) $\frac{\partial u}{\partial x}(0,y) = 0$, $u(L,y) = g(y)$, $u(x,0) = 0$, $u(x,H) = 0$

(d) $u(0,y) = g(y)$, $u(L,y) = 0$, $\frac{\partial u}{\partial y}(x,0) = 0$, $u(x,H) = 0$

(e) $u(0,y) = 0$, $u(L,y) = 0$, $u(x,0) - \frac{\partial u}{\partial y}(x,0) = 0$, $u(x,H) = f(x)$

(f) $u(0,y) = f(y)$, $u(L,y) = 0$, $\frac{\partial u}{\partial y}(x,0) = 0$, $\frac{\partial u}{\partial y}(x,H) = 0$

(g) $\frac{\partial u}{\partial x}(0,y) = 0$, $\frac{\partial u}{\partial x}(L,y) = 0$, $u(x,0) = \begin{cases} 0 & x > L/2 \\ 1 & x < L/2 \end{cases}$, $\frac{\partial u}{\partial y}(x,H) = 0$

(h) $u(0,y) = 0$, $u(L,y) = g(y)$, $u(x,0) = 0$, $u(x,H) = 0$

For this specific class of problem – Sturm-Liouville – over the given intervals, we will omit consideration of negative eigenvalues, and simply investigate the cases of $\lambda \geq 0$.

(a) Since the boundary conditions are homogeneous in x, we have

$$u(x,y) = \phi(x)\,h(y)$$

Then we write the Laplacian as

$$\nabla^2 u = h(y)\,\frac{d^2\phi}{dx^2} + \phi(x)\,\frac{d^2 h}{dy^2} = 0$$

Separating variables and setting equal to the separation constant, derive

$$\frac{1}{\phi(x)}\frac{d^2\phi}{dx^2} = -\frac{1}{h(y)}\frac{d^2 h}{dy^2} = -\lambda$$

where we choose the negative sign to ensure that we have oscillation in x (to satisfy the two homogeneous boundary conditions) and exponentials in y. Now, to solve the eigenvalue problem, first consider the positive case:

<u>$\lambda > 0$:</u> The general spatial solution is

$$\phi(x) = c_1 \cos\left(\sqrt{\lambda}x\right) + c_2 \sin\left(\sqrt{\lambda}x\right)$$

Imposing the boundary conditions, derive

$$\frac{d\phi}{dx}(0) = c_2\sqrt{\lambda} = 0 \implies c_2 = 0$$
$$\frac{d\phi}{dx}(L) = -c_1\sqrt{\lambda}\sin\left(\sqrt{\lambda}L\right) = 0$$

where in the second boundary condition, a non-trivial solution corresponds to eigenvalues

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2$$

with eigenfunctions

$$\phi(x) = \cos\left(\frac{n\pi x}{L}\right)$$

$\underline{\lambda = 0}$: The general spatial solution in this case is

$$\phi(x) = c_1 + c_2 x$$

The boundary conditions are simply

$$\frac{d\phi}{dx}(0) = \frac{d\phi}{dx}(L) = c_2 = 0$$

Leaving us with $\phi(x) = c_1$.

Now we move onto the y-equations. For the positive eigenvalues λ_n we derived, we have general solution

$\underline{\lambda > 0}$:

$$h(y) = a_1\cosh\left(\frac{n\pi y}{L}\right) + a_2\sinh\left(\frac{n\pi y}{L}\right)$$

where the boundary conditions produce

$$h(0) = a_1 = 0$$
$$h(H) = a_2\sinh\left(\frac{n\pi H}{L}\right)$$

which yields corresponding eigenfunction

$$h(y) = \sinh\left(\frac{n\pi y}{L}\right)$$

Lastly, we examine the zero eigenvalue case.

$\underline{\lambda = 0}$:

$$h(y) = a_1 + a_2 y$$

with boundary conditions

$$h(0) = a_1 = 0$$
$$h(H) = a_2 H$$

leaving us with eigenfunction

$$h(y) = a_2 y$$

We summarize our equations in the table below. For the following problems, I will simply list the eigenfunctions derived from the boundary conditions in such tables for the sake of brevity, unless special considerations warrant a more detailed derivation.

	$\lambda = 0$	$\lambda = \left(\frac{n\pi}{L}\right)^2 > 0$
$\phi(x)$	c_1	$\cos\left(\frac{n\pi x}{L}\right)$
$h(y)$	$a_2 y$	$\sinh\left(\frac{n\pi y}{L}\right)$

Combining solutions and using superposition, the full temperature distribution is

$$u(x,y) = A_0 y + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)\sinh\left(\frac{n\pi y}{L}\right)$$

To identify the coefficients we rely on the non-homogeneous boundary condition: $u(x,H) = f(x)$. At that point, we have

$$f(x) = u(x,H) = A_0 H + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)\sinh\left(\frac{n\pi H}{L}\right)$$

First we consider the case when $n > 0$, i.e. for the positive eigenvalues. Here is the key point: **we must project our boundary condition $f(x)$ onto each of the directions of the Hilbert space spanned by the orthogonal eigenbasis corresponding to the non-zero eigenvalues. The coefficients A_m correspond to the "coordinates" of $f(x)$ as projected on that specific direction in the Hilbert space**. By orthogonality, multiply all terms by $\cos(m\pi x/L)$ and integrate from 0 to L to get

$$\begin{aligned}\int_0^L f(x)\cos\left(\frac{m\pi x}{L}\right)\,dx = \int_0^L & A_0 H \cos\left(\frac{m\pi x}{L}\right)\,dx \\ &+ \int_0^L A_m \cos^2\left(\frac{m\pi x}{L}\right)\sinh\left(\frac{m\pi H}{L}\right)\,dx\end{aligned}$$

This in turn allows us to compute A_m:

$$A_m = \frac{2}{L\sinh(m\pi H/L)}\int_0^L f(x)\cos\left(\frac{m\pi x}{L}\right)\,dx$$

Lastly, to compute A_0, we analogously project the boundary condition onto the eigenfunctions corresponding to the zero eigenvalue. Truly, we deal with only a singular eigenfunction: the constant, which we can reduce to 1. Another way to conceptualize this is accounting for all of the basis vectors of the Hilbert space in x

$$\left\{\cos\left(\frac{n\pi x}{L}\right)\,\middle|\, n \in \mathbb{N}\cup\{0\}\right\}$$

where in the $n > 0$ case we considered the subspace spanned by

$$\{\cos(n\pi x/L) \mid n > 0\}$$

and in the $n = 0$ case we account for the basis vector $\{1\}$. With that out of the way, calculate

$$\int_0^L f(x)\,dx = \int_0^L A_0 H\,dx + \int_0^L \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) \sinh\left(\frac{n\pi H}{L}\right)\,dx$$

Observe that most conveniently, the entirety of the second term vanishes due to the orthogonality condition in the cosine function against the constant eigenfunction 1. This leaves us with

$$A_0 = \frac{1}{HL}\int_0^L f(x)\,dx$$

(b) With boundary conditions homogeneous in y, we have

$$u(x, y) = h(x)\,\phi(y)$$

The Laplacian is then

$$\nabla^2 u = \phi(y)\,\frac{d^2 h}{dx^2} + h(x)\,\frac{d^2\phi}{dy^2} = 0$$

Separating variables and setting equal to the separation constant, derive

$$\frac{1}{h(x)}\frac{d^2 h}{dx^2} = -\frac{1}{\phi(y)}\frac{d^2\phi}{dy^2} = \lambda$$

where the positive sign is chosen to ensure oscillation in y and exponentials in x. The eigenfunctions and corresponding eigenvalues are

	$\lambda = \left(\frac{n\pi}{H}\right)^2 > 0$
$\phi(y)$	$\sin\left(\frac{n\pi y}{H}\right)$
$h(x)$	$\sinh\left(\frac{n\pi}{H}(x - L)\right)$

By superposition, we have temperature distribution

$$u(x, y) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi y}{H}\right) \sinh\left(\frac{n\pi}{H}(x - L)\right)$$

To derive the coefficients A_n, first differentiate with respect to x and examine the point $(0, y)$, where the non-homogeneous condition is realized

$$\frac{\partial u}{\partial x}(0, y) = \sum_{n=1}^{\infty} A_n \left(\frac{n\pi}{H}\right) \sin\left(\frac{n\pi y}{H}\right) \cosh\left(-\frac{n\pi L}{H}\right) = g(y)$$

Then multiplying the right equality by $\sin(m\pi y/H)$, integrating from 0 to H, and isolating the coefficient produces

$$A_m = \frac{2}{m\pi \cosh(m\pi L/H)} \int_0^H g(y) \sin\left(\frac{m\pi y}{H}\right)\,dy$$

where we appeal to the evenness of the hyperbolic cosine function to change the argument sign to positive.

(c) Once again we are homogeneous in y, so we define

$$u(x,y) = h(x)\,\phi(y)$$

with corresponding Laplacian

$$\nabla^2 u = \phi(y)\,\frac{d^2h}{dx^2} + h(x)\,\frac{d^2\phi}{dy^2} = 0$$

Separating variables and setting equal to the separation constant, derive

$$\frac{1}{h(x)}\frac{d^2h}{dx^2} = -\frac{1}{\phi(y)}\frac{d^2\phi}{dy^2} = \lambda$$

where the separation constant sign is chosen to ensure oscillation in y. The eigenfunctions and corresponding eigenvalues are

	$\lambda_n = \left(\frac{n\pi}{H}\right)^2 > 0$
$\phi(y)$	$\sin\left(\frac{n\pi y}{H}\right)$
$h(x)$	$\cosh\left(\frac{n\pi x}{H}\right)$

By superposition, we have complete temperature distribution

$$u(x,y) = \sum_{n=1}^{\infty} A_n \cosh\left(\frac{n\pi x}{H}\right)\sin\left(\frac{n\pi y}{H}\right)$$

where A_n is determined by evaluating at (L,y):

$$u(L,y) = \sum_{n=1}^{\infty} A_n \cosh\left(\frac{n\pi L}{H}\right)\sin\left(\frac{n\pi y}{H}\right) = g(y)$$

and then multiplying by $\sin(m\pi y/H)$ and integrating from 0 to H to derive

$$A_m = \frac{2}{H\cosh(m\pi L/H)}\int_0^H g(y)\sin\left(\frac{m\pi y}{H}\right)\,dy$$

(d) Homogeneity of the boundary conditions in y produces

$$u(x,y) = h(x)\,\phi(y)$$

The Laplacian is

$$\nabla^2 u = \phi(y)\,\frac{d^2h}{dx^2} + h(x)\,\frac{d^2\phi}{dy^2} = 0$$

Separating variables and setting equal to the separation constant, derive

$$\frac{1}{h(x)}\frac{d^2h}{dx^2} = -\frac{1}{\phi(y)}\frac{d^2\phi}{dy^2} = \lambda$$

where the sign of the separation constant ensures oscillation in y and exponentials in x. The eigenfunctions and corresponding eigenvalues are

	$\lambda_n = \left[\dfrac{(2n-1)\,\pi}{2H}\right]^2 > 0$
$\phi(y)$	$\cos\left[\dfrac{(2n-1)\,\pi y}{2H}\right]$
$h(x)$	$\sinh\left[\dfrac{(2n-1)\,\pi(x-L)}{2H}\right]$

By superposition, we have temperature distribution

$$u(x,y) = \sum_{n=1}^{\infty} A_n \sinh\left[\frac{(2n-1)\,\pi(x-L)}{2H}\right]\cos\left[\frac{(2n-1)\,\pi y}{2H}\right]$$

where the Fourier coefficient is derived as:

$$A_m = -\frac{2}{H\sinh[(2m-1)\,\pi L/2H]}\int_0^H g(y)\cos\left[\frac{(2m-1)\,\pi y}{2H}\right]\,dy$$

(e) We switch to homogeneity of the boundary conditions in x, so we define

$$u(x,y) = \phi(x)\,h(y)$$

which has Laplacian

$$\nabla^2 u = h(y)\,\frac{d^2\phi}{dx^2} + \phi(x)\,\frac{d^2 h}{dy^2} = 0$$

Separating variables and setting equal to the separation constant, derive

$$\frac{1}{\phi(x)}\frac{d^2\phi}{dx^2} = -\frac{1}{h(y)}\frac{d^2 h}{dy^2} = -\lambda$$

which guarantees oscillation in x and exponentials in y. The eigenfunctions and corresponding eigenvalues are

	$\lambda_n = \left(\dfrac{n\pi}{L}\right)^2$
$\phi(x)$	$\sin\left(\dfrac{n\pi x}{L}\right)$
$h(y)$	$\left(\dfrac{n\pi}{L}\right)\cosh\left(\dfrac{n\pi y}{L}\right) + \sinh\left(\dfrac{n\pi y}{L}\right)$

By superposition, we have temperature distribution

$$u(x,y) = \sum_{n=1}^{\infty} A_n h_n(y)\sin\left(\frac{n\pi x}{L}\right)$$

where

$$h_n(y) = \left(\frac{n\pi}{L}\right)\cosh\left(\frac{n\pi y}{L}\right) + \sinh\left(\frac{n\pi y}{L}\right)$$

and the Fourier coefficient is derived as

$$A_m = \frac{2}{Lh_m(H)}\int_0^L f(x)\sin\left(\frac{m\pi x}{L}\right)\,dx$$

(f) Homogeneity of the boundary conditions in y yields

$$u(x,y) = h(x)\,\phi(y)$$

with corresponding Laplacian

$$\nabla^2 u = \phi(y)\,\frac{d^2h}{dx^2} + h(x)\,\frac{d^2\phi}{dy^2} = 0$$

Separating variables and setting equal to the separation constant, derive

$$\frac{1}{h(x)}\frac{d^2h}{dx^2} = -\frac{1}{\phi(y)}\frac{d^2\phi}{dy^2} = \lambda$$

which ensures oscillation in y and exponentials in x. The eigenfunctions and corresponding eigenvalues are

	$\lambda_0 = 0$	$\lambda_n = \left(\frac{n\pi}{H}\right)^2 > 0$
$\phi(y)$	c	$\cos\left(\frac{n\pi y}{H}\right)$
$h(x)$	$1 - \frac{x}{L}$	$\sinh\left[\frac{n\pi}{H}(x-L)\right]$

By superposition, we have solution

$$u(x,y) = A_0\left(1 - \frac{x}{L}\right) + \sum_{n=1}^{\infty} A_n \sinh\left[\frac{n\pi}{H}(x-L)\right]\cos\left(\frac{n\pi y}{H}\right)$$

where the Fourier coefficients are given by

$$A_m = \begin{cases} \dfrac{1}{H}\displaystyle\int_0^H f(y)\;dy, & m = 0 \\ -\dfrac{2}{H\sinh(m\pi L/H)}\displaystyle\int_0^H f(y)\cos\left(\frac{m\pi y}{H}\right)\;dy, & m > 0 \end{cases}$$

(g) Our boundary conditions are homogeneous in x, which gives us

$$u(x,y) = \phi(x)\,h(y)$$

which has corresponding Laplacian

$$\nabla^2 u = h(y)\,\frac{d^2\phi}{dx^2} + \phi(x)\,\frac{d^2h}{dy^2} = 0$$

Separating variables and setting equal to the separation constant, find

$$\frac{1}{\phi(x)}\frac{d^2\phi}{dx^2} = -\frac{1}{h(y)}\frac{d^2h}{dy^2} = -\lambda$$

which guarantees oscillation in x and exponentials in y. The eigenfunctions and corresponding eigenvalues are

	$\lambda_0 = 0$	$\lambda_n = \left(\frac{n\pi}{L}\right)^2 > 0$
$\phi(x)$	c_1	$\cos\left(\frac{n\pi x}{L}\right)$
$h(y)$	a_1	$\cosh\left[\frac{n\pi}{L}(y-H)\right]$

which combine and with superposition produce

$$u(x,y) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)\cosh\left[\frac{n\pi}{L}(y-H)\right]$$

where

$$A_0 = \frac{1}{2}, \quad A_m = \frac{2}{m\pi\cosh(m\pi H/L)}\sin\left(\frac{m\pi}{2}\right)$$

Note that the derivation proceeds analogously to the problem posed in (a), where the presence of non-trivial eigenfunctions corresponding to $\lambda_0 = 0$ indicates two different projections of the boundary condition onto the eigenbases associated with the zero and non-zero eigenvalues.

(h) We end this first set of exercises with homogeneity of the boundary conditions in y, giving us form

$$u(x,y) = h(x)\,\phi(y)$$

The corresponding Laplacian is

$$\nabla^2 u = \phi(y)\,\frac{d^2h}{dx^2} + h(x)\,\frac{d^2\phi}{dy^2} = 0$$

Separating variables and setting equal to the separation constant, derive

$$\frac{1}{h(x)}\frac{d^2h}{dx^2} = -\frac{1}{\phi(y)}\frac{d^2\phi}{dy^2} = \lambda$$

which ensures oscillation in y and exponentials in x. The eigenfunctions and corresponding eigenvalues are

	$\lambda_n = \left(\frac{n\pi}{H}\right)^2 > 0$
$\phi(y)$	$\sin\left(\frac{n\pi y}{H}\right)$
$h(x)$	$\sinh\left(\frac{n\pi x}{H}\right)$

By superposition, we have temperature distribution

$$u(x,y) = \sum_{n=1}^{\infty} A_n \sinh\left(\frac{n\pi x}{H}\right)\sin\left(\frac{n\pi y}{H}\right)$$

with Fourier coefficient

$$A_m = \frac{2}{H\sinh(m\pi L/H)} \int_0^H g(y)\sin\left(\frac{m\pi y}{H}\right)\, dy$$

Question: 2.5.2

Consider $u(x, y)$ satisfying Laplace's equation inside a rectangle ($0 < x < L$, $0 < y < H$) subject to the boundary conditions

$$\begin{aligned} \frac{\partial u}{\partial x}(0, y) &= 0, & \frac{\partial u}{\partial y}(x, 0) &= 0 \\ \frac{\partial u}{\partial x}(L, y) &= 0, & \frac{\partial u}{\partial y}(x, H) &= f(x) \end{aligned}$$

(a) *Without* solving this problem, briefly explain the physical condition under which there is a solution to this problem.

(b) Solve this problem by the method of separation of variables. Show that the method works only under the condition of part (a). [*Hint*: You may use (2.5.16) without derivation.]

(c) The solution [part (b)] has an arbitrary constant. Determine it by consideration of the time-dependent heat equation (1.5.11) subject to the initial condition

$$u(x, y, 0) = g(x, y)$$

(a) Conceptually, net heat flow through the boundary must be zero in order for the steady-state to exist, namely the solvability condition

$$\oint \nabla u \cdot \hat{\boldsymbol{n}}\, ds = \int_0^L \frac{\partial u}{\partial y}(x, H)\ dx = \int_0^L f(x)\ dx = 0$$

(b) The boundary conditions are homogeneous in x, so we separate variables as

$$u(x, y) = \phi(x)\, h(y)$$

The corresponding Laplacian at equilibrium is

$$\nabla^2 u = h(y)\,\frac{d^2\phi}{dx^2} + \phi(x)\,\frac{d^2 h}{dy^2} = 0$$

And separating variables and setting equal to the separation constant produces

$$\frac{1}{\phi(x)}\frac{d^2\phi}{dx^2} = -\frac{1}{h(y)}\frac{d^2 h}{dy^2} = -\lambda$$

which guarantees oscillation in x and exponentials in y. The eigenfunctions and corresponding eigenvalues are

	$\lambda_0 = 0$	$\lambda_n = \left(\frac{n\pi}{L}\right)^2 > 0$
$\phi(x)$	1	$\cos\left(\frac{n\pi x}{L}\right)$
$h(y)$	1	$\cosh\left(\frac{n\pi y}{L}\right)$

By superposition, we have temperature distribution

$$u(x, y) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) \cosh\left(\frac{n\pi y}{L}\right)$$

where the Fourier coefficients for the eigenbases with non-zero eigenvalues are given by

$$A_m = \frac{2}{m\pi \sinh(m\pi H/L)} \int_0^L f(x) \cos\left(\frac{m\pi x}{L}\right)\, dx$$

However, in the case of multiplying through by the constant eigenfunction $\phi(x) = 1$ and integrating, we find

$$\begin{aligned} \int_0^L \frac{\partial u}{\partial y}(x, H)\, dx &= \sum_{n=1}^{\infty} A_n \left(\frac{n\pi}{L}\right) \sinh\left(\frac{n\pi H}{L}\right) \int_0^L \cos\left(\frac{n\pi x}{L}\right)\, dx \\ &= \int_0^L f(x)\, dx \\ &= 0 \end{aligned}$$

Which is consistent with the solvability condition in part (a). The other insight here is that A_0 *is undetermined from the boundary conditions*, and in equilibrium, *any* value of A_0 is consistent with all four boundary conditions.

(c) Where we do determine A_0 is from the initial condition. First, recall that one plan of approach for the time-dependent case is to compute the rate of change of the total energy, namely

$$\begin{aligned} \frac{\partial E}{\partial t} = \iint c\rho \frac{\partial u}{\partial t}\, dA &= K_0 \iint \nabla \cdot \nabla u\, dA \\ &= \oint \nabla u \cdot \hat{\boldsymbol{n}}\, ds \\ &= \int_0^L f(x)\, dx \\ &= 0 \end{aligned}$$

where we use the divergence theorem and the solvability condition in the second and third lines, respectively. We have now established that **energy is conserved**.

What next? This implies

$$\iint u(x,y,0)\; dA = \iint u(x,y)\; dA = \iint g(x,y)\; dA$$

which conceptually translates to initial energy equals to energy at equilibrium. We may simply integrate

$$\begin{aligned}\int_0^H \int_0^L \left[A_0 + \sum_{n=1}^{\infty} A_n \cos\!\left(\frac{n\pi x}{L}\right)\cosh\!\left(\frac{n\pi y}{L}\right)\right] dx\, dy &= A_0 HL \\ &= \iint g(x,y)\; dA\end{aligned}$$

implying

$$A_0 = \frac{1}{HL}\iint g(x,y)\; dA$$

Observe that physically, the transient modes carry net-zero energy, which is the interpretation of the entire summation vanishing by the area integral.

Question: 2.5.3

Solve Laplace's equation *outside* a circular disk ($r \geq a$) subject to the boundary condition [*Hint*: In polar coordinates,

$$\nabla^2 u = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0$$

it is known that if $u(r,\theta) = \phi(\theta)\, G(r)$, then $\frac{r}{G}\frac{d}{dr}\left(r\frac{dG}{dr}\right) = -\frac{1}{\phi}\frac{d^2\phi}{d\theta^2}$.]:

(a) $u(a,\theta) = \ln(2) + 4\cos(3\theta)$

(b) $u(a,\theta) = f(\theta)$

You may assume that $u(r,\theta)$ remains finite as $r \to \infty$.

By separable variables, let

$$u(r,\theta) = \phi(\theta)\, G(r)$$

Which produces the equation

$$\frac{r}{G}\frac{d}{dr}\left(r\frac{dG}{dr}\right) = -\frac{1}{\phi}\frac{d^2\phi}{d\theta^2} = \lambda$$

First assume $\lambda > 0$ and examine the angular eqution. The periodicity conditions

$$\begin{aligned}\phi(-\pi) &= \phi(\pi) \\ \frac{d\phi}{d\theta}(-\pi) &= \frac{d\phi}{d\theta}(\pi)\end{aligned}$$

must be satisfied. Since we have oscillation in θ, derive general form

$$\phi(\theta) = c_1 \cos\!\left(\sqrt{\lambda}\theta\right) + c_2 \sin\!\left(\sqrt{\lambda}\theta\right)$$

Enforcing the first boundary condition produces

$$\phi(-\pi) = c_1 \cos\left(\sqrt{\lambda}\pi\right) - c_2 \sin\left(\sqrt{\lambda}\pi\right) = c_1 \cos\left(\sqrt{\lambda}\pi\right) + c_2 \sin\left(\sqrt{\lambda}\pi\right) = \phi(\pi)$$

implying

$$2c_2 \sin\left(\sqrt{\lambda}\pi\right) = 0$$

The second:

$$\begin{aligned}\frac{d\phi}{d\theta}(-\pi) &= c_1\sqrt{\lambda}\sin\left(\sqrt{\lambda}\pi\right) + c_2\sqrt{\lambda}\cos\left(\sqrt{\lambda}\pi\right)\\ \frac{d\phi}{d\theta}(\pi) &= -c_1\sqrt{\lambda}\sin\left(\sqrt{\lambda}\pi\right) + c_2\sqrt{\lambda}\cos\left(\sqrt{\lambda}\pi\right)\end{aligned}$$

implying

$$2c_1\sqrt{\lambda}\sin\left(\sqrt{\lambda}\pi\right) = 0$$

Combined, the boundary conditions suggest that either $c_1 = c_2 = 0$ or $\lambda = n^2$. Since we desire a non-trivial solution, this is what produces our eigenvalues.

From the radial equation, we seek to solve

$$\frac{r}{G}\frac{d}{dr}\left(r\frac{dG}{dr}\right) = n^2$$

where, when rearranging and evaluating terms, we have the equidimensional equation

$$r^2\frac{d^2G}{dr^2} + r\frac{dG}{dr} - n^2G = 0$$

Which is solved by supposing $G(r) = r^p$, solving for p, and deriving $p = \pm n$. Therefore the general radial solution is

$$G(r) = a_1r^n + a_2r^{-n}$$

Now, here is where some physical intuition comes in. We are considering the area *beyond* $r = a$. Originally, we had to guarantee that $|G(0)| < \infty$; that is, we bound $G(r)$ at the origin to be finite. In this case, since we consider the distribution of heat far from the origin, it is reasonable to suggest that the boundary condition in this case be

$$|G(\infty)| < \infty$$

Imposing this boundary condition implies

$$G(\infty) = a_1r^n + a_2r^{-n} < \infty \quad \Longrightarrow \quad a_1 = 0$$

Repeat the process for $\lambda = 0$. For the angular equation, derive

$$\phi(\theta) = c_1 + c_2\theta$$

where the periodicity conditions suggest $c_2 = 0$, leaving us with the constant eigenfunction. In the radial equation, we are left to solve

$$\frac{d}{dr}\left(r\frac{dG}{dr}\right) = 0$$

which has solution

$$G(r) = b_1 + b_2 \ln(r)$$

Similarly, we apply the limiting condition $|G(\infty)| < \infty$, forcing $b_2 = 0$, and leaving us once again with the constant eigenfunction. The complete set of eigenfunctions and corresponding eigenvalues is thus

	$\lambda_0 = 0$	$\lambda_n = n^2 > 0$
$\phi(\theta)$	1	$\sin(n\theta)$, $\cos(n\theta)$
$G(r)$	1	r^{-n}

Therefore, combining the radial and angular solutions and appealing to superposition gives us general temperature distribution

$$u(r,\theta) = \sum_{n=0}^{\infty} A_n r^{-n}\cos(n\theta) + \sum_{n=1}^{\infty} B_n r^{-n}\sin(n\theta)$$

(a)

$$\begin{aligned} u(a,\theta) &= \sum_{n=0}^{\infty} A_n a^{-n}\cos(n\theta) + \sum_{n=1}^{\infty} B_n a^{-n}\sin(n\theta) \\ &= \ln(2) + 4\cos(3\theta) \end{aligned}$$

implies

$$A_0 = \ln(2)\,, \quad A_3 = 4a^3, \quad A_{n\neq 0,3} = 0, \quad B_n = 0$$

(b)

$$u(a,\theta) = \sum_{n=0}^{\infty} A_n a^{-n}\cos(n\theta) + \sum_{n=1}^{\infty} B_n a^{-n}\sin(n\theta) = f(\theta)$$

To calculate the Fourier coefficients, go case-by-case. For A_0, multiply through by $\phi(\theta) = 1$ and integrate from $-\pi$ to π to get

$$A_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(\theta)\ d\theta$$

For A_n and B_n, use orthogonality and integrate over the angular bounds to get

$$\begin{aligned} A_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\cos(n\theta)\ d\theta \\ B_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\sin(n\theta)\ d\theta \end{aligned}$$

Question: 2.5.4

For Laplace's equation inside a circular disk ($r \leq a$), using (2.5.45) and (2.5.47), show that

$$u(r,\theta) = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\overline{\theta})\left[-\frac{1}{2} + \sum_{n=0}^{\infty}\left(\frac{r}{a}\right)^n \cos\left(n\left(\theta - \overline{\theta}\right)\right)\right] d\overline{\theta}$$

Using $\cos(z) = \mathrm{Re}\left[e^{iz}\right]$, sum the resulting geometric series to obtain Poisson's integral formula.

Using the formulae for the Fourier coefficients, substitute into the solution to the polar Laplacian. Note that we define a dummy variable of integration $\overline{\theta}$ for the coefficient formulae:

$$u(r,\theta) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(\overline{\theta})\ d\overline{\theta} + \sum_{n=1}^{\infty}\frac{1}{\pi}\left(\frac{r}{a}\right)^n \cos(n\theta)\int_{-\pi}^{\pi} f(\overline{\theta})\cos\left(n\overline{\theta}\right)\ d\overline{\theta}$$
$$+\sum_{n=1}^{\infty}\frac{1}{\pi}\left(\frac{r}{a}\right)^n \sin(n\theta)\int_{-\pi}^{\pi} f(\overline{\theta})\sin\left(n\overline{\theta}\right)\ d\overline{\theta}$$

Now, the advantage of defining this dummy variable is to be able to move the θ functions inside of the integrals, and exploiting the angle sum and difference trigonometric identities to simplify. Do this and combine terms to derive

$$u(r,\theta) = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\overline{\theta})\left[\frac{1}{2} + \sum_{n=1}^{\infty}\left(\frac{r}{a}\right)^n\left(\cos(n\theta)\cos\left(n\overline{\theta}\right) + \sin(n\theta)\sin\left(n\overline{\theta}\right)\right)\right] d\overline{\theta}$$

The morass of trigonometic terms can be combined into $\cos\left(n\left(\theta - \overline{\theta}\right)\right)$, giving us

$$u(r,\theta) = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\overline{\theta})\left[\frac{1}{2} + \sum_{n=1}^{\infty}\left(\frac{r}{a}\right)^n \cos\left(n\left(\theta - \overline{\theta}\right)\right)\right] d\overline{\theta}$$

And lastly, adding one to the second of the bracketed terms and subtracting from the first allows us to readjust the summation index to commence from $n = 0$, as desired:

$$u(r,\theta) = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\overline{\theta})\left[-\frac{1}{2} + \sum_{n=0}^{\infty}\left(\frac{r}{a}\right)^n \cos\left(n\left(\theta - \overline{\theta}\right)\right)\right] d\overline{\theta}$$

To derive Poisson's integral formula, first isolate the summation and rewrite as

$$\sum_{n=0}^{\infty}\left(\frac{r}{a}\right)^n \cos\left(n\left(\theta - \overline{\theta}\right)\right) = \mathrm{Re}\left[\sum_{n=0}^{\infty}\left(\frac{re^{in\left(\theta - \overline{\theta}\right)}}{a}\right)^n\right]$$

Since the modulus $|r/a| < 1$, we apply the geometric series formula to derive

$$\sum_{n=0}^{\infty}\left(\frac{re^{in\left(\theta - \overline{\theta}\right)}}{a}\right)^n = \frac{1}{1 - \frac{r}{a}e^{i\left(\theta - \overline{\theta}\right)}}$$

Now, the complex conjugate of the denominator is

$$1 - \frac{r}{a} e^{-i\left(\theta - \overline{\theta}\right)}$$

Multiplying the numerator and denominator with this complex conjugate and explicitly writing out the trigonometric functions, calculate the *real* part as

$$\frac{a^2 - ar\cos\left(\theta - \overline{\theta}\right)}{a^2 + r^2 - 2ar\cos\left(\theta - \overline{\theta}\right)}$$

We still have the $-1/2$ term to deal with. Summing together, find

$$-\frac{1}{2} + \frac{a^2 - ar\cos\left(\theta - \overline{\theta}\right)}{a^2 + r^2 - 2ar\cos\left(\theta - \overline{\theta}\right)} = \frac{a^2 - r^2}{2\left[a^2 + r^2 - 2ar\cos\left(\theta - \overline{\theta}\right)\right]}$$

For the finale, look back to $u(r, \theta)$. Replace the bracketed term with our result to derive Poisson's integral formula:

$$u(r, \theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f\left(\overline{\theta}\right) \frac{a^2 - r^2}{a^2 + r^2 - 2ar\cos\left(\theta - \overline{\theta}\right)}\, d\overline{\theta}$$

Question: 2.5.5

Solve Laplace's equation inside the quarter-circle of radius 1 ($0 \le \theta \le \pi/2$, $0 \le r \le 1$) subject to the boundary conditions [*Hint*: In polar coordinates,

$$\nabla^2 u = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0$$

it is known that if $u(r, \theta) = \phi(\theta)\, G(r)$, then $\frac{r}{G}\frac{d}{dr}\left(r\frac{dG}{dr}\right) = -\frac{1}{\phi}\frac{d^2\phi}{d\theta^2}$.]:

(a) $\frac{\partial u}{\partial \theta}(r, 0) = 0, \quad u\left(r, \frac{\pi}{2}\right) = 0, \quad u(1, \theta) = f(\theta)$

(b) $\frac{\partial u}{\partial \theta}(r, 0) = 0, \quad \frac{\partial u}{\partial \theta}\left(r, \frac{\pi}{2} = 0\right), \quad u(1, \theta) = f(\theta)$

(c) $u(r, 0) = 0, \quad u\left(r, \frac{\pi}{2}\right) = 0, \quad \frac{\partial u}{\partial r}(1, \theta) = f(\theta)$

(d) $\frac{\partial u}{\partial \theta}(r, 0) = 0, \quad \frac{\partial u}{\partial \theta}\left(r, \frac{\pi}{2}\right) = 0, \quad \frac{\partial u}{\partial r}(1, \theta) = g(\theta)$

Show that the solution [part (d)] exists only if $\int_0^{\pi/2} g(\theta)\, d\theta = 0$. Explain this condition physically.

(a) Since the boundary conditions in θ are homogeneous, we set separation constant

$$\frac{r}{G}\frac{d}{dr}\left(r\frac{dG}{dr}\right) = -\frac{1}{\phi}\frac{d^2\phi}{d\theta^2} = \lambda$$

<u>$\lambda > 0$:</u>

Our angular equation is

$$\phi(\theta) = c_1 \cos\left(\sqrt{\lambda}\theta\right) + c_2 \sin\left(\sqrt{\lambda}\theta\right)$$

which is subject to the boundary conditions

$$\frac{d\phi}{d\theta}(0) = c_2\sqrt{\lambda} = 0 \qquad \Longrightarrow \qquad c_2 = 0$$
$$\phi\left(\frac{\pi}{2}\right) = c_1 \cos\left(\frac{\sqrt{\lambda}\pi}{2}\right) = 0$$

We derive eigenvalues

$$\lambda_n = (2n-1)^2, \quad n = 1, 2, ...$$

which gives angular eigenfunction

$$\phi(\theta) = \cos[(2n-1)\,\theta]$$

For the radial equation, we end up with the equidimensional equation

$$r^2\frac{d^2G}{dr^2} + r\frac{dG}{dr} - (2n-1)^2\,G = 0$$

Substitute $G = r^p$ and solve the characteristic polynomial

$$p(p-1) + p - (2n-1)^2 = 0 \qquad \Longrightarrow \qquad p = \pm(2n-1)$$

Leaving us with radial equation

$$G(r) = a_1 r^p + a_2 r^{-p}$$

Now, we must impose the bounding condition that $|G(0) < \infty|$. This forces $a_2 = 0$, as otherwise G would diverge to infinity. Then our radial eigenfunction is

$$G(r) = r^{2n-1}$$

<u>$\lambda = 0$:</u>

We have angular equation

$$\phi(\theta) = c_1 + c_2\theta$$

with boundary conditions

$$\frac{d\phi}{d\theta}(0) = c_2 = 0$$
$$\phi\left(\frac{\pi}{2}\right) = c_1 = 0$$

meaning no mode can exist for $n = 0$. We summarize our eigenfunctions and corresponding eigenvalues as

	$\lambda_n = (2n-1)^2 > 0$
$\phi(\theta)$	$\cos[(2n-1)\,\theta]$
$G(r)$	r^{2n-1}

By superposition, we have temperature distribution

$$u(r,\theta) = \sum_{n=1}^{\infty} A_n r^{2n-1} \cos[(2n-1)\,\theta]$$

where A_n is given by first setting

$$u(1,\theta) = \sum_{n=1}^{\infty} A_n \cos[(2n-1)\,\theta] = f(\theta)$$

and then appealing to orthogonality of the eigenfunctions to derive

$$A_m = \frac{\int_0^{\pi/2} f(\theta)\cos[(2m-1)\,\theta]\;d\theta}{\int_0^{\pi/2} \cos^2[(2m-1)\,\theta]\;d\theta} = \frac{4}{\pi}\int_0^{\pi/2} f(\theta)\cos[(2m-1)\,\theta]\;d\theta$$

(b) With boundary conditions homogeneous in θ, write

$$\frac{r}{G}\frac{d}{dr}\left(r\frac{dG}{dr}\right) = -\frac{1}{\phi}\frac{d^2\phi}{d\theta^2} = \lambda$$

which produces eigenfunctions with corresponding eigenvalues

	$\lambda_0 = 0$	$\lambda_n = (2n)^2 > 0$
$\phi(\theta)$	1	$\cos(2n\theta)$
$G(r)$	1	r^{2n}

By superposition, the temperature distribution is

$$u(r,\theta) = \sum_{n=0}^{\infty} A_n r^{2n} \cos(2n\theta)$$

with Fourier coefficients

$$A_0 = \frac{2}{\pi}\int_0^{\pi/2} f(\theta)\;d\theta$$
$$A_m = \frac{4}{\pi}\int_0^{\pi/2} f(\theta)\cos(2m\theta)\;d\theta$$

(c) Given that the boundary conditions in θ are homogeneous, we set separation constant

$$\frac{r}{G}\frac{d}{dr}\left(r\frac{dG}{dr}\right) = -\frac{1}{\phi}\frac{d^2\phi}{d\theta^2} = \lambda$$

which produce eigenfunctions and corresponding eigenvalues

	$\lambda_n = (2n)^2 > 0$
$\phi(\theta)$	$\sin(2n\theta)$
$G(r)$	r^{2n}

By superposition, we have temperature distribution

$$u(r, \theta) = \sum_{n=1}^{\infty} A_n r^{2n} \sin(2n\theta)$$

with Fourier coefficients

$$A_m = \frac{2}{m\pi} \int_0^{\pi/2} f(\theta) \sin(2m\theta)\ d\theta$$

(d) Homogeneity in θ from the boundary conditions enables us to set separation constant

$$\frac{r}{G} \frac{d}{dr} \left(r \frac{dG}{dr} \right) = -\frac{1}{\phi} \frac{d^2\phi}{d\theta^2} = \lambda$$

and since the boundary conditions in θ are exactly as those of part (b), we have temperature distribution

$$u(r, \theta) = \sum_{n=0}^{\infty} A_n r^{2n} \cos(2n\theta)$$

For A_n, $n > 0$, we appeal to cosine orthogonality to derive

$$A_m = \frac{2}{m\pi} \int_0^{\pi/2} g(\theta) \cos(2m\theta)\ d\theta$$

In the case when we multiply through by the constant eigenfunction and integrate, however, we find

$$\int_0^{\pi/2} \frac{\partial u}{\partial r}(1, \theta)\ d\theta = \int_0^{\pi/2} \sum_{n=1}^{\infty} 2A_n n \cos(2n\theta)\ d\theta = \int_0^{\pi/2} g(\theta)\ d\theta = 0$$

Which forces our solvability condition, or that net flux is zero. Note that A_0 is undetermined (as in the case of exercise 2.5.2); it will remain so unless we have an initial condition to determine it.

Question: 2.5.6

Solve Laplace's equation inside a semicircle of radius a ($0 < r < a$, $0 < \theta < \pi$) subject to the boundary conditions [*Hint*: In polar coordinates,

$$\nabla^2 u = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0$$

it is known that if $u(r,\theta) = \phi(\theta)\,G(r)$, then $\frac{r}{G}\frac{d}{dr}\left(r\frac{dG}{dr}\right) = -\frac{1}{\phi}\frac{d^2\phi}{d\theta^2}$.]:

(a) $u = 0$ on the diameter and $u(a,\theta) = g(\theta)$

(b) the diameter is insulated and $u(a,\theta) = g(\theta)$

(a) The two boundary conditions in θ are

$$u(r,0) = u(r,\pi) = 0$$

The eigenfunctions and corresponding eigenvalues are

	$\lambda_n = n^2 > 0$
$\phi(\theta)$	$\sin(n\theta)$
$G(r)$	r^n

By superposition, the temperature distribution is

$$u(r,\theta) = \sum_{n=1}^{\infty} A_n r^n \sin(n\theta)$$

where the Fourier coefficients are

$$A_m a^m = \frac{2}{\pi}\int_0^{\pi} g(\theta)\sin(m\theta)\;d\theta$$

(b) Now, the two boundary conditions in θ are

$$\frac{\partial u}{\partial \theta}(r,0) = \frac{\partial u}{\partial \theta}(r,\pi) = 0$$

The eigenfunctions and eigenvalues switch to

	$\lambda_0 = 0$	$\lambda_n = n^2 > 0$
$\phi(\theta)$	1	$\cos(n\theta)$
$G(r)$	1	r^n

And the temperature distribution superposes to become

$$u(r,\theta) = A_0 + \sum_{n=1}^{\infty} A_n r^n \cos(n\theta)$$

with Fourier coefficients

$$\begin{aligned} A_0 &= \frac{1}{\pi}\int_0^{\pi} g(\theta)\ d\theta \\ A_m a^m &= \frac{2}{\pi}\int_0^{\pi} g(\theta)\cos(m\theta)\ d\theta \end{aligned}$$

Question: 2.5.11

Do Exercise 1.5.8.

See solution manual for Chapter 1.

Question: 2.5.12

(a) Using the divergence theorem, determine an alternative expression for $\iiint u\nabla^2 u\, dx\, dy\, dz$.

(b) Using part (a), prove that the solution of Laplace's equation $\nabla^2 u = 0$ (with u given on the boundary) is unique.

(c) Modify part (b) if $\nabla u \cdot \hat{\boldsymbol{n}} = 0$ on the boundary.

(d) Modify part (b) if $\nabla u \cdot \hat{\boldsymbol{n}} + hu = 0$ on the boundary. Show that Newton's law of cooling corresponds to $h > 0$.

(a) The strategy is to equate the integrand to some form of $\nabla \cdot A$:

$$u\nabla^2 u = \nabla \cdot A$$

To do so, observe that

$$\nabla \cdot (u\nabla u) = |\nabla u|^2 + u\nabla^2 u$$

Implying $u\nabla^2 u = \nabla \cdot (u\nabla u) - |\nabla u|^2$. By the divergence theorem,

$$\begin{aligned}\iiint_R u\nabla^2 u \, dx \, dy \, dz &= \iiint_R \left(\nabla \cdot (u\nabla u) - |\nabla u|^2\right) \, dx \, dy \, dz \\ &= \oiint u\nabla u \cdot \hat{\boldsymbol{n}} \, dS - \iiint_R |\nabla u|^2 \, dV\end{aligned}$$

which is called **Green's first identity**.

(b) Suppose both u and v solve Laplace's equation and satisfy the same boundary condition. Define $w \equiv u - v$. By superposition,

$$\nabla^2 w = \nabla^2 u - \nabla^2 v = 0$$

which enables us to write

$$\oiint w\nabla w \cdot \hat{\boldsymbol{n}} \, dS - \iiint_R |\nabla w|^2 \, dV = 0$$

On the boundary, we know that $u = v$ as they satisfy the same boundary condition, so it must be the case that $w = 0$ and the surface integral vanishes. This leaves us with

$$\iiint_R |\nabla w|^2 \, dV = 0$$

which forces $|\nabla w|^2 = 0$ and consequently, $\nabla w = 0$. Therefore w is a constant. But since we have determined that $w = 0$ on the boundary, it must be true that $w = 0$ *everywhere*, proving $u = v$.

(c) In particular, we change the boundary condition to $\nabla u \cdot \hat{\boldsymbol{n}} = 0$, and suppose another solution v also satisfies that boundary condition. Similarly define $w \equiv u - v$. Then it must follow by linearity that $\nabla w \cdot \hat{\boldsymbol{n}} = 0$, and we have

$$\iiint_R |\nabla w|^2 \, dV = 0$$

Ergo, once more, $\nabla w = 0$. We can deduce that w is at most a constant. Unlike in part (b), we have insufficient information to ascertain if $w = 0$, so we leave our conclusion at that.

(d) Assume solutions u and v and consider w as above. Our new boundary condition produces

$$\oiint hw^2 \, dS + \iiint_R |\nabla w|^2 \, dV = 0$$

Now, suppose $h > 0$. Then it must be that $hw^2 \geq 0$ and additionally, we know $|\nabla w|^2 \geq 0$. Now, since both integrands are non-negative but their sum must be zero, equality persists only when **both integrands are zero**, requiring $w = 0$ on the boundary. Since we also have that $\nabla w = 0$, then we know w is constant, proving that $w = 0$ everywhere.[1] In

[1] Alternatively, by the maximum principle, $w = 0$ *everywhere* inside the volume, and $u = v$ everywhere.

the case where $h = 0$, part (d) reduces to part (c), and for $h < 0$, we can lay no claim on even the constancy of w, for the two integrals can simply balance each other out (assuming non-zero w).

In the statement of Newton's law of cooling:

$$-\nabla u \cdot \hat{\boldsymbol{n}} = h(u - u_b)$$

where u is the temperature at the surface, u_b the outside temperature, and $h = H/K_0$, we can intuitively see that $u > u_b$ corresponds to a cooler outside temperature than on the surface. Then it follows that $-K_0 \nabla u \cdot \hat{\boldsymbol{n}} > 0$. The sign across the equality stays consistent only when $h > 0$.

Question: 2.5.13

Prove that the temperature satisfying Laplace's equation cannot attain its minimum in the interior.

If u is non-constant and a solution to Laplace's equation, then so is $-u$. From the maximum principle, if $-u$ cannot attain its maximum value in the interior, then u cannot attain its minimum value in the interior. Physically, suppose u was the minimum temperature in the interior at a point P. Then by diffusion, the temperature at point P would rise, and cease to be the minimum, contradicting the steady-stateness of P.

Question: 2.5.14

Show that the "backward" heat equation

$$\frac{\partial u}{\partial t} = -k \frac{\partial^2 u}{\partial x^2}$$

subject to $u(0, t) = u(L, t) = 0$ and $u(x, 0) = f(x)$, is *not* well posed. [*Hint:* Show that if the data are changed an arbitrarily small amount, for example,

$$f(x) \to f(x) + \frac{1}{n} \sin\left(\frac{n\pi x}{L}\right)$$

for large n, then the solution $u(x, t)$ changes by a large amount.]

The solution to the backwards heat equation becomes

$$u(x, t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{k(n\pi/L)^2 t}$$

where

$$B_m = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{m\pi x}{L}\right) \, dx$$

The strategy here is to appeal to linearity and subtract from $u(x, t)$ the perturbed solution, $u^*(x, t)$. Consider the coefficients generated by the perturbed initial condition B_m^*:

$$B_m^* = \frac{2}{L} \int_0^L \left[f(x) + \frac{1}{n} \sin\left(\frac{n\pi x}{L}\right) \right] \sin\left(\frac{m\pi x}{L}\right) \, dx$$

Taking the difference $B_m - B_m^*$, compute

$$B_m - B_m^* = -\frac{2}{L}\int_0^L \frac{1}{n}\sin\left(\frac{m\pi x}{L}\right)\sin\left(\frac{n\pi x}{L}\right)\,dx$$
$$= \begin{cases} -\frac{1}{n}, & m = n \\ 0, & m \neq n \end{cases}$$

Substituting into our difference of solutions (letting $m = n$, with terms for which $m \neq n$ vanishing by orthogonality), find

$$u(x,t) - u^*(x,t) = -\frac{1}{n}\sin\left(\frac{n\pi x}{L}\right)e^{k(n\pi/L)^2 t}$$

Fix t. As $n \to \infty$, the exponential vastly outpaces the $1/n$ term, demonstrating that the difference between the perturbed and unperturbed solutions also goes to infinity, ascertaining that the backwards heat equation is not well posed.

Question: 2.5.15

Solve Laplace's equation inside a semi-infinite strip ($0 < x < \infty$, $0 < y < H$) subject to the boundary conditions [*Hint*: In Cartesian coordinates, $\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, inside a semi-infinite strip ($0 \le y \le H$ and $0 \le x < \infty$), it is known that if $u(x,y) = F(x)\,G(y)$, then $\frac{1}{F}\frac{d^2F}{dx^2} = -\frac{1}{G}\frac{d^2G}{dy^2}$.]:

(a) $\frac{\partial u}{\partial y}(x,0) = 0, \quad \frac{\partial u}{\partial y}(x,H) = 0, \quad u(0,y) = f(y)$

(b) $u(x,0) = 0, \quad u(x,H) = 0, \quad u(0,y) = f(y)$

(c) $u(x,0) = 0, \quad u(x,H) = 0, \quad \frac{\partial u}{\partial x}(0,y) = f(y)$

(d) $\frac{\partial u}{\partial y}(x,0) = 0, \quad \frac{\partial u}{\partial y}(x,H) = 0, \quad \frac{\partial u}{\partial x}(0,y) = f(y)$

Show that the solution [part (d)] exists only if $\int_0^H f(y)\,dy = 0$.

For each problem, it is implied that the boundary condition as $x \to \infty$ must be $|F(\infty)| < \infty$.

(a) We wish for our eigenfunctions to oscillate in y and have exponentials in x, so from the definition of $u(x,y)$ in the problem statement and the separation of variables thereafter, we set equal to

$$\frac{1}{F}\frac{d^2F}{dx^2} = -\frac{1}{G}\frac{d^2G}{dy^2} = \lambda$$

Our eigenfunctions and corresponding eigenvalues are

	$\lambda = 0$	$\lambda = \left(\frac{n\pi}{H}\right)^2 > 0$
$F(x)$	1	$e^{-n\pi x/H}$
$G(y)$	1	$\cos\left(\frac{n\pi y}{H}\right)$

Observe that in the case of $F(x)$, we have basis functions spanning the solution space, $e^{\sqrt{\lambda_n}x}$ and $e^{-\sqrt{\lambda_n}x}$. The boundedness condition forces $e^{\sqrt{\lambda_n}x}$ to drop out of the solution. We use the exponentials as the basis functions rather than the hyperbolic sine and cosine, as both diverge with x, so neither alone can satisfy the boundedness condition. The complete solution is given by

$$u(x, y) = A_0 + \sum_{n=1}^{\infty} A_n e^{-n\pi x/H} \cos\left(\frac{n\pi y}{H}\right)$$

with coefficients given by

$$A_n = \begin{cases} \dfrac{1}{H}\displaystyle\int_0^H f(y)\ dy, & n = 0 \\ \dfrac{2}{H}\displaystyle\int_0^H f(y)\cos\left(\frac{n\pi y}{H}\right)\ dy, & n \geq 1 \end{cases}$$

Observe that $\lim_{x\to\infty} u(x, y) = A_0$, which is simply the mean of the boundary condition $f(y)$.

(b) Once again, we have oscillation in y, so we set the sign of the eigenvalue as

$$\frac{1}{F}\frac{d^2F}{dx^2} = -\frac{1}{G}\frac{d^2G}{dy^2} = \lambda$$

Our eigenfunctions to our eigenvalue are

	$\lambda_n = \left(\frac{n\pi}{H}\right)^2 > 0$
$F(x)$	$e^{-n\pi x/H}$
$G(y)$	$\sin\left(\frac{n\pi y}{H}\right)$

which form complete solution

$$u(x, y) = \sum_{n=1}^{\infty} A_n e^{-n\pi x/H} \sin\left(\frac{n\pi y}{H}\right)$$

with coefficients

$$A_n = \frac{2}{H}\int_0^H f(y)\sin\left(\frac{n\pi y}{H}\right)\ dy, \quad n \geq 1$$

Notice that we have no A_0, as when $\lambda = 0$, we receive only the trivial solution, so it turns out that $\lambda = 0$ is not an eigenvalue at all. Thus $n = 0$ fails to be a mode in this problem.

(c) Barring the non-homogeneous boundary condition, all other boundary conditions from part (b) persist here, leaving us with complete solution

$$u(x, y) = \sum_{n=1}^{\infty} A_n e^{-n\pi x/H} \sin\left(\frac{n\pi y}{H}\right)$$

The only difference is in differentiating $u(x, y)$, setting equal to the boundary condition:

$$\frac{\partial u}{\partial x}(0, y) = \sum_{n=1}^{\infty} A_n \left(-\frac{n\pi}{H}\right) \sin\left(\frac{n\pi y}{H}\right) = f(y)$$

and appealing to orthogonality to integrate and deriving

$$A_n = -\frac{2}{n\pi} \int_0^H f(y) \sin\left(\frac{n\pi y}{H}\right)\, dy$$

(d) Similarly, since the boundary conditions in y and the boundedness condition in x apply here as in part (a), we retain solution

$$u(x, y) = A_0 + \sum_{n=1}^{\infty} A_n e^{-n\pi x/H} \cos\left(\frac{n\pi y}{H}\right)$$

Differentiating and applying the boundary condition, we find

$$\frac{\partial u}{\partial x}(0, y) = \sum_{n=1}^{\infty} A_n \left(-\frac{n\pi}{H}\right) \cos\left(\frac{n\pi y}{H}\right) = f(y)$$

which gives coefficients

$$A_n = -\frac{2}{n\pi} \int_0^H f(y) \cos\left(\frac{n\pi y}{H}\right)\, dy$$

Several interesting aspects to note here. One, we see that A_0 is annihilated completely in the differentiation. As such, A_0 faces no constraint, and any arbitrary value would satisfy all of the boundary conditions.

Moreover, observe that the boundary conditions in y leave the top and bottom of the strip completely insulated. Physically, we can see that there could not be any heat flow *out* of the strip at the far end, as the modes of $u(x, y)$ exponentially decay as $x \to \infty$. Therefore, we must mandate that the **net heat flow through the boundary at $x = 0$ must be zero for a steady state to exist**, for otherwise, we would not have a net zero change in energy in the strip. Thus the requirement

$$\int_0^H f(y)\, dy = 0$$

in which case A_0 takes on an arbitrary value, and otherwise no solution exists.

Question: 2.5.16

Consider Laplace's equation inside a rectangle $0 \le x \le L$, $0 \le y \le H$, with the boundary conditions

$$\frac{\partial u}{\partial x}(0, y) = 0, \quad \frac{\partial u}{\partial x}(L, y) = g(y)\,, \quad \frac{\partial u}{\partial y}(x, 0) = 0, \quad \frac{\partial u}{\partial y}(x, H) = f(x)$$

(a) What is the solvability condition and its physical interpretation?

(b) Show that $u(x, y) = A\left(x^2 - y^2\right)$ is a solution if $f(x)$ and $g(y)$ are constants [under the conditions of part (a)].

(c) Under the conditions of part (a), solve the general case [nonconstant $f(x)$ and $g(y)$]. *Hints*: Use part (b) and the fact that $f(x) = f_{\text{av}} + [f(x) - f_{\text{av}}]$, where $f_{\text{av}} = \frac{1}{L}\int_0^L f(x)\ dx$.]

(a) The solvability condition is

$$\int_0^H g(y)\ dy + \int_0^L f(x)\ dx = 0$$

which physically ensures that there is zero net heat flux across the boundary, and thereby permitting the steady state to exist.

(b) Let $f(x) \equiv f$ and $g(y) \equiv g$ be constants, then

$$\int_0^H g\,dy + \int_0^L f\,dx = gH + fL = 0$$

First, ensure that our solution solves Laplace's equation:

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2A - 2A = 0$$

Second, the boundary conditions are satisfied only if:

$$\begin{aligned} \frac{\partial u}{\partial x}(0, y) &= 2A \cdot 0 &&= 0 \\ \frac{\partial u}{\partial x}(L, y) &= 2A \cdot L &&= g \\ \frac{\partial u}{\partial y}(x, 0) &= -2A \cdot 0 &&= 0 \\ \frac{\partial u}{\partial y}(x, H) &= -2A \cdot H &&= f \end{aligned}$$

Now, from the boundary conditions, we can derive two expressions for A:

$$A = \frac{g}{2L}, \quad A = -\frac{f}{2H}$$

The test is to determine if these are, in fact, describing the same A. Our strategy arises from rewriting our solvability condition by subtracting fL and dividing by $2HL$:

$$\frac{g}{2L} = -\frac{f}{2H}$$

which precisely demonstrates uniqueness of A, and thus uniqueness of $u(x, y)$ up to an additive constant.

(c) The key insight here is to apply superposition: **instead of dealing with all nonhomogeneous boundary conditions simultaneously, we can separate the problem so that we only have to deal with one at a time**.

In the first case, let $\partial u/\partial x = 0$ at $x = 0$ and $x = L$. The boundary conditions in y are now

$$\frac{\partial u}{\partial y}(x,0) = 0$$
$$\frac{\partial u}{\partial y}(x,H) = f(x) - f_{\text{av}}$$

Our **solvability condition**, which expresses the physical condition of net zero heat flux across the boundary, then becomes

$$\int_0^L [f(x) - f_{\text{av}}]\, dx = 0$$

Then our eigenvalue problem leads to oscillation in x, and we can derive

$$u_f(x,y) = A_0 + \sum_{n=1}^{\infty} A_n \cos\Big(\frac{n\pi x}{L}\Big) \cosh\Big(\frac{n\pi y}{L}\Big)$$

Differentiating in y, we find

$$\frac{\partial u_f}{\partial y}(x,y) = \sum_{n=1}^{\infty} A_n \Big(\frac{n\pi}{L}\Big) \cos\Big(\frac{n\pi x}{L}\Big) \sinh\Big(\frac{n\pi y}{L}\Big)$$

and evaluate

$$\frac{\partial u_f}{\partial y}(x,0) = 0$$
$$\frac{\partial u_f}{\partial y}(x,H) = \sum_{n=1}^{\infty} A_n \Big(\frac{n\pi}{L}\Big) \cos\Big(\frac{n\pi x}{L}\Big) \sinh\bigg(\frac{n\pi H}{L}\bigg) = f(x) - f_{\text{av}}$$

Then by orthogonality, in the case of $n \geq 1$, the coefficients are

$$A_n = \frac{2}{n\pi \sinh(n\pi H/L)} \int_0^L [f(x) - f_{\text{av}}] \cos\Big(\frac{n\pi x}{L}\Big)\, dx$$

And in the case when $n = 0$, we see that A_0 can be any arbitrary constant, and we observe the consistency of the solvability condition:

$$\frac{1}{L}\int_0^L [f(x) - f_{\text{av}}]\, dx = f_{\text{av}} - f_{\text{av}} = 0$$

Analogously, solve the $g(y)$ piece which has boundary conditions

$$\begin{aligned}
\frac{\partial u}{\partial x}(0, y) &= 0 \\
\frac{\partial u}{\partial x}(L, y) &= g(y) - g_{\text{av}} \\
\frac{\partial u}{\partial y}(x, 0) &= 0 \\
\frac{\partial u}{\partial y}(x, H) &= 0
\end{aligned}$$

which has solution

$$u_g(x, y) = B_0 + \sum_{n=1}^{\infty} B_n \cos\left(\frac{n\pi y}{H}\right) \cosh\left(\frac{n\pi x}{H}\right)$$

where B_0 is an arbitrary constant and

$$B_n = \frac{2}{n\pi \sinh(n\pi L/H)} \int_0^H [g(y) - g_{\text{av}}] \cos\left(\frac{n\pi y}{H}\right) dy$$

Now using the solution from part (b) in the case of constant f and g, we may sum the solutions to find our final answer:

$$\begin{aligned}
u(x, y) &= u_b + u_f + u_g \\
&= A\left(x^2 - y^2\right) + C + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) \cosh\left(\frac{n\pi y}{L}\right) \\
&\quad + \sum_{n=1}^{\infty} B_n \cos\left(\frac{n\pi y}{H}\right) \cosh\left(\frac{n\pi x}{H}\right)
\end{aligned}$$

where $C = A_0 + B_0$ is itself an arbitrary constant, and from the solvability condition in part (a) and the derivation of A in part (b), we need only set the constants $f = f_{\text{av}}$ and $g = g_{\text{av}}$ to get $A = g_{\text{av}}/2L = -f_{\text{av}}/2L$.

Question: 2.5.17

Show that the mass density $\rho(x, t)$ satisfies $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0$ due to conservation of mass.

Express conservation of mass as

$$M \equiv \int_R \rho(x, t) \, dV$$

where R is a fixed volume. Recall our derivation of the heat equation in which we balanced the rate of change of energy with the total flux and sum of internal sources (eq. 1.5.1). **This is the exact same principle, but with mass**. Conceptually, the rate of change of mass in a volume is determined by the mass flux density, or mass density advected in and out of the boundary. Derive (note there are no sources by conservation)

$$\frac{d}{dt} \int_R \rho \, dV = \int_R \frac{\partial \rho}{\partial t} \, dV = -\oint (\rho \boldsymbol{u}) \cdot \hat{\boldsymbol{n}} \, dS$$

From the divergence theorem, the surface integral becomes

$$-\oint (\rho \boldsymbol{u}) \cdot \hat{\boldsymbol{n}}\, dS = -\int_R \nabla \cdot (\rho \boldsymbol{u})\ dV$$

which allows us to balance the equation as

$$\int_R \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u})\right]\ dV = 0$$

and since this equality holds for any arbitrary region R, it follows that the integrand itself is zero, producing

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0$$

Our equation is the famous **continuity equation**.

Question: 2.5.18

If the mass density is constant, using the result of Exercise 2.5.17, show that $\nabla \cdot \boldsymbol{u} = 0$.

For constant ρ, we have $\partial\rho/\partial t = 0$ and $\nabla\rho = \mathbf{0}$. Then

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = \rho(\nabla \cdot \boldsymbol{u}) + \boldsymbol{u} \cdot \nabla\rho = \rho(\nabla \cdot \boldsymbol{u}) = 0$$

implying that $\nabla \cdot \boldsymbol{u} = 0$ as $\rho > 0$, always.

Question: 2.5.19

Show that the streamlines are parallel to the fluid velocity.

Definitionally, streamlines (ψ = constant) are level curves. We know that the gradient is perpendicular to level curves; to prove that the fluid velocity is parallel to the streamlines, we can equivalently demonstrate that

$$\boldsymbol{u} \cdot \nabla\psi = 0$$

Define $\boldsymbol{u} \equiv (u, v)$. Since $\nabla\psi = (\partial\psi/\partial x, \partial\psi/\partial y)$, we can use the definitions of the stream functions (Haberman eq. 2.5.49)

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}$$

and compute our dot product as

$$\boldsymbol{u} \cdot \nabla\psi = \frac{\partial \psi}{\partial y}\frac{\partial \psi}{\partial x} - \frac{\partial \psi}{\partial x}\frac{\partial \psi}{\partial y} = 0$$

establishing orthogonality between $\boldsymbol{u}$ and $\nabla\psi$, and thus parallelism between $\boldsymbol{u}$ and the level curves of ψ.

Question: 2.5.20

Show that anytime there is a stream function, $\nabla \times \boldsymbol{u} = -\hat{\boldsymbol{k}}\nabla^2\psi$.

Define $\boldsymbol{u} \equiv (u, v, 0) = (\partial\psi/\partial y, -\partial\psi/\partial x, 0)$, where there is no vertical component to otherwise planar flow, and the u and v components are independent of z. Compute

$$\begin{aligned}
\nabla \times \boldsymbol{u} &= \begin{vmatrix} \hat{\boldsymbol{i}} & \hat{\boldsymbol{j}} & \hat{\boldsymbol{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial\psi}{\partial y} & -\frac{\partial\psi}{\partial x} & 0 \end{vmatrix} \\
&= -\hat{\boldsymbol{i}}\frac{\partial v}{\partial z} + \hat{\boldsymbol{j}}\frac{\partial u}{\partial z} - \hat{\boldsymbol{k}}\left(\frac{\partial}{\partial x}\left(\frac{\partial\psi}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial\psi}{\partial y}\right)\right) \\
&= -\hat{\boldsymbol{k}}\left(\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2}\right) \\
&= -\hat{\boldsymbol{k}}\nabla^2\psi
\end{aligned}$$

Question: 2.5.21

From $u = \frac{\partial\psi}{\partial y}$ and $v = -\frac{\partial\psi}{\partial x}$, derive $u_r = \frac{1}{r}\frac{\partial\psi}{\partial\theta}$, $u_\theta = -\frac{\partial\psi}{\partial r}$.

In two-dimensional polar coordinates, we rotate our u,v-coordinates by θ. We can represent this matrix rotation as

$$\begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} u\cos(\theta) + v\sin(\theta) \\ -u\sin(\theta) + v\cos(\theta) \end{bmatrix} = \begin{bmatrix} u_r \\ u_\theta \end{bmatrix}$$

Next, from the chain rule and the coordinate conversions $x = r\cos(\theta)$ and $y = r\sin(\theta)$, we achieve the desired results:

$$\begin{aligned}
\frac{\partial\psi}{\partial\theta} &= \frac{\partial\psi}{\partial x}\frac{\partial x}{\partial\theta} + \frac{\partial\psi}{\partial y}\frac{\partial y}{\partial\theta} \\
&= -\frac{\partial\psi}{\partial x} r\sin(\theta) + \frac{\partial\psi}{\partial y} r\cos(\theta) \\
&= r[v\sin(\theta) + u\cos(\theta)] \\
&= ru_r \\
\frac{\partial\psi}{\partial r} &= \frac{\partial\psi}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial\psi}{\partial y}\frac{\partial y}{\partial r} \\
&= \frac{\partial\psi}{\partial x}\cos(\theta) + \frac{\partial\psi}{\partial y}\sin(\theta) \\
&= -v\cos(\theta) + u\sin(\theta) \\
&= -u_\theta
\end{aligned}$$

remembering that $u = \partial\psi/\partial y$ and $v = -\partial\psi/\partial x$.

Question: 2.5.22

Show the drag force is zero for a uniform flow past a cylinder including circulation.

The x-component of the force (drag) is

$$F_x = \frac{1}{2}\rho \int_0^{2\pi} \left[-\frac{c_1}{r} - U\left(1 + \frac{a^2}{r^2}\right)\sin(\theta)\right]^2 \cos(\theta)\, a\, d\theta$$

At the surface of the cylinder, where the drag force is applied, we have $r = a$. This simplifies the expression to be

$$F_x = \frac{1}{2}\rho \int_0^{2\pi} \left[-\frac{c_1}{a} - 2U\sin(\theta)\right]^2 \cos(\theta)\, a\, d\theta$$

Solving and appealing to the symmetries of the integrand, we derive

$$F_x = \frac{1}{2}\rho \int_0^{2\pi} \left[\frac{c_1^2}{a^2} + \frac{4c_1 U}{a}\sin(\theta) + 4U^2\sin^2(\theta)\right] \cos(\theta)\, a\, d\theta = 0$$

which we can quickly discern by observing that the integrals of $\cos(\theta)$, $\sin(\theta)\cos(\theta)$, and $\sin^2(\theta)\cos(\theta)$ all vanish over $[0, 2\pi]$.

Question: 2.5.23

Consider the velocity u_θ at the cylinder. Where do the maximum and minimum occur?

We have at $r = a$

$$u_\theta(a, \theta) = -\frac{c_1}{a} - 2U\sin(\theta)$$

Optimizing, deduce

$$\frac{du_\theta}{d\theta} = -2U\cos(\theta) = 0 \quad \Longleftrightarrow \quad \theta = \frac{\pi}{2}, \frac{3\pi}{2}$$

As for where the minimum and maximum velocities occur, we have a series of cases that are dependent on the sign of U. The table below conveys the corresponding angles for each:

	$U > 0$	$U < 0$
Max of u_θ	$\theta = \frac{3\pi}{2}$	$\theta = \frac{\pi}{2}$
Min of u_θ	$\theta = \frac{\pi}{2}$	$\theta = \frac{3\pi}{2}$

Question: 2.5.24

Consider the velocity u_θ at the cylinder. If the circulation is negative, show that the velocity will be larger above the cylinder than below.

With negative circulation, we have $c_1 > 0$ (ensuring $-2\pi c_1 < 0$) in

$$u_\theta = -\frac{c_1}{a} - 2U\sin(\theta)$$

Assuming we also have $U > 0$, $|u_\theta(\pi/2)| > |u_\theta(3\pi/2)|$. Therefore velocity is larger above the cylinder than below.

Question: 2.5.25

A stagnation point is a place where $\boldsymbol{u} = 0$. For what values of the circulation does a stagnation point exist on the cylinder?

On the cylinder, the r-component evaluates to

$$u_r(a,\theta) = U\left(1 - \frac{a^2}{a^2}\right)\cos(\theta) = 0$$

So now it remains to determine when u_θ is zero:

$$u_\theta(a,\theta) = -\frac{c_1}{a} - 2U\sin(\theta) = 0$$

for which equality is satisfied when

$$\sin(\theta) = -\frac{c_1}{2aU}$$

We know that $0 \leq |\sin(\theta)| \leq 1$, meaning that a solution will only exist if

$$0 \leq \left|\frac{c_1}{2aU}\right| \leq 1$$

Using our expression for circulation

$$\int_0^{2\pi} u_\theta r\, d\theta = -2\pi c_1$$

we substitute for c_1 and conclude

$$0 \leq \frac{1}{4\pi a}\left|\frac{1}{U}\int_0^{2\pi} u_\theta r\, d\theta\right| \leq 1$$

Question: 2.5.26

For what values of θ will $u_r = 0$ off the cylinder? For these θ, where (for what values of r) will $u_\theta = 0$ also?

Assume negative circulation with $c_1, U > 0$. Our task is to find θ such that

$$u_r = U\left(1 - \frac{a^2}{r^2}\right)\cos(\theta) = 0$$

for $r \neq a$. Immediately we can discern that these θ are $\frac{\pi}{2}$ and $\frac{3\pi}{2}$. The two expressions we get for u_θ and their corresponding quadratic and roots are

	$\theta = \pi/2$	$\theta = 3\pi/2$
$u_\theta(r,\theta) = 0$	$-\frac{c_1}{r} - U\left(1 + \frac{a^2}{r^2}\right) = 0$	$-\frac{c_1}{r} + U\left(1 + \frac{a^2}{r^2}\right) = 0$
Quadratic	$r^2 + \frac{c_1}{U}r + a^2 = 0$	$r^2 - \frac{c_1}{U}r + a^2 = 0$
Roots	$\frac{-c_1/U \pm \sqrt{c_1^2/U^2 - 4a^2}}{2}$	$\frac{c_1/U \pm \sqrt{c_1^2/U^2 - 4a^2}}{2}$

Now, two constraints we must impose are for r to be real and positive. The first constraint follows only when the discriminant satisfies

$$1 \leq \left(\frac{c_1}{2aU}\right)^2$$

which, when compared to the inequality established in the previous problem, is the *complement* range of values for $|c_1/2aU|$.

In the second constraint, we observe that for $\theta = \pi/2$, since $c_1/U > \sqrt{c_1^2/U^2 - 4a^2}$, the roots corresponding to that quadratic *are never positive*. Therefore we must reject the $\theta = \pi/2$ solution altogether. This leaves us with $\theta = 3\pi/2$.

Lastly, we have two roots – two stagnation points. We care about the stagnation point that is *off* the cylinder, namely for $r > a$. Using the inequality derived from the discriminant, one direct way we can ascertain which root satisfies this condition is by observing that

$$a^2 \leq \left(\frac{c_1}{2U}\right)^2 \quad \Longrightarrow \quad a \leq \frac{c_1}{2U}$$

which we can do since all terms are positive. Now, it is apparent that

$$\frac{c_1}{2U} \leq \frac{c_1}{2U} + \frac{1}{2}\sqrt{\left(\frac{c_1}{U}\right)^2 - 4a^2}$$

which conveys to us that the first root is the stagnation point that falls outside of the cylinder, and therefore the value of r we have been searching for. The other lies inside of the cylinder, and can be ignored.

In the special case of equality, namely where

$$a = \frac{c_1}{2U}$$

then the point lies on the cylinder itself.

Question: 2.5.27

Show that $\psi = \alpha\frac{\sin(\theta)}{r}$ satisfies Laplace's equation. Show that the streamlines are circles. Graph the streamlines.

The task is to demonstrate that

$$\nabla^2\psi = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2\psi}{\partial\theta^2} = 0$$

From the derivatives

$$\frac{\partial\psi}{\partial r} = -\alpha\frac{\sin(\theta)}{r^2}, \quad \frac{\partial}{\partial r}\left(r\frac{\partial\psi}{\partial r}\right) = \frac{\partial}{\partial r}\left(-\alpha\frac{\sin(\theta)}{r}\right) = \alpha\frac{\sin(\theta)}{r^2}, \quad \frac{\partial^2\psi}{\partial\theta^2} = -\alpha\frac{\sin(\theta)}{r}$$

we can substitute and conclude

$$\frac{1}{r}\left(\alpha\frac{\sin(\theta)}{r^2}\right) - \frac{1}{r^2}\left(\alpha\frac{\sin(\theta)}{r}\right) = 0$$

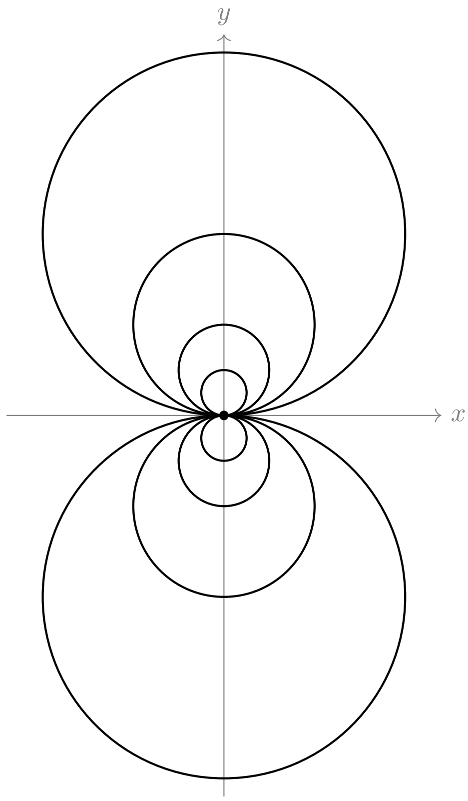
To ascertain that ψ is a circle, first we set equal to a constant: call it C. Multiply through by r^2 and convert to Cartesian to derive

$$\alpha r\sin(\theta) = Cr^2 \qquad \Longrightarrow \qquad \alpha y = C\big(x^2+y^2\big)$$

Through some algebraic manipulation and completing the square, we have the circle equation

$$x^2 + \left(y - \frac{\alpha}{2C}\right)^2 = \frac{\alpha^2}{4C^2}$$

with radius $R = \alpha/2C$ and center $(0, R)$. Below are graphs of a family of these circles, with $\alpha = 1$ and $R = \pm 1/4$, $\pm 1/2$, ± 1, and ± 2.



Question: 2.5.28

Solve Laplace's equation inside a rectangle:

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

subject to the boundary conditions

$$\begin{aligned} u(0,y) &= g(y) \qquad & u(x,0) &= 0 \\ u(L,y) &= 0 \qquad & u(x,H) &= 0 \end{aligned}$$

Let $u(x,y) = \phi(x)\,G(y)$. Substitute into Laplace's equation to find

$$G(y)\,\frac{d^2\phi}{dx^2} + \phi(x)\,\frac{d^2 G}{dy^2} = 0$$

Divide through by $\phi(x)\,G(y)$ and subtract the y term to find

$$\frac{1}{\phi(x)}\frac{d^2\phi}{dx^2} = -\frac{1}{G(y)}\frac{d^2 G}{dy^2}$$

Since our boundary conditions are homogeneous in y, we want the eigenfunctions in that variable to oscillate, so we set both x and y terms equal to λ:

$$\frac{1}{\phi(x)}\frac{d^2\phi}{dx^2} = -\frac{1}{G(y)}\frac{d^2 G}{dy^2} = \lambda$$

Suppose that $\lambda > 0$. Solve the y equation first. It is a simple second-order ordinary differential equation solved by

$$G(y) = c_1 \cos\left(\sqrt{\lambda} y\right) + c_2 \sin\left(\sqrt{\lambda} y\right)$$

Apply the boundary conditions in y. We find

$$\begin{aligned} G(0) &= c_1 = 0 \\ G(H) &= c_2 \sin\left(\sqrt{\lambda} H\right) = 0 \end{aligned}$$

and for a nontrivial solution to emerge, we can require

$$\lambda_n = \left(\frac{n\pi}{H}\right)^2, \quad n = 1, 2, ...$$

which comprise our eigenvalues for this eigenfunction problem. Speaking of which, for $\lambda > 0$, we have corresponding eigenfunction

$$G(y) = \sin\left(\frac{n\pi y}{H}\right)$$

In the case where $\lambda = 0$, integrate the second derivative twice to find solution

$$G(y) = c_1 + c_2 y$$

Apply boundary conditions to see that only the trivial solution exists:

$$\begin{aligned} G(0) &= c_1 = 0 \\ G(H) &= c_2 H = 0 \quad \implies \quad c_2 = 0 \end{aligned}$$

And finally, suppose $\lambda < 0$. Then define $s = -\lambda$, which is positive. Solve

$$\frac{1}{G(y)} \frac{d^2 G}{dy^2} = s$$

which has solution

$$G(y) = c_1 \cosh\left(\sqrt{s} y\right) + c_2 \sinh\left(\sqrt{s} y\right)$$

Applying the boundary conditions yields

$$\begin{aligned} G(0) &= c_1 = 0 \\ G(H) &= c_2 \sinh\left(\sqrt{s} H\right) = 0 \quad \implies \quad c_2 = 0 \end{aligned}$$

which merely produces the trivial solution yet again, enabling us to conclude that in this case, we only have positive eigenvalues. Next we solve the x problem with

$$\phi(x) = a_1 \cosh\left[\frac{n\pi}{H}(x - L)\right] + a_2 \sinh\left[\frac{n\pi}{H}(x - L)\right]$$

Impose the zero boundary condition in x to derive our eigenfunction

$$\phi(L) = a_1 = 0 \quad \implies \quad \phi(x) = \sinh\left[\frac{n\pi}{H}(x - L)\right]$$

Reassembling the product $\phi(x)\,G(y)$ and taking the linear combination of all possible solutions, we have

$$u(x,y)=\sum_{n=1}^{\infty}A_n\sin\left(\frac{n\pi y}{H}\right)\sinh\left[\frac{n\pi}{H}(x-L)\right]$$

To determine the coefficients A_n, evaluate at $x=0$ and multiply both sides by $\sin(m\pi y/H)$ to find

$$u(0,y)=\sum_{n=1}^{\infty}A_n\sin\left(\frac{n\pi y}{H}\right)\sin\left(\frac{m\pi y}{H}\right)\sinh\left[\frac{n\pi}{H}(-L)\right]=g(y)\sin\left(\frac{m\pi y}{H}\right)$$

Integrate both sides over $[0,H]$, and by orthogonality, all $n\neq m$ terms will be eliminated, leaving us with

$$A_m\sinh\left[\frac{m\pi}{H}(-L)\right]\underbrace{\int_0^H\sin^2\left(\frac{m\pi y}{H}\right)\,dy}_{H/2}=\int_0^H g(y)\sin\left(\frac{m\pi y}{H}\right)\,dy$$

Lastly, isolate A_m to get

$$A_m=\frac{2}{H\sinh(m\pi(-L)/H)}\int_0^H g(y)\sin\left(\frac{m\pi y}{H}\right)\,dy$$

Question: 2.5.29

Solve Laplace's equation inside a circle of radius a,

$$\nabla^2u=\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2u}{\partial\theta^2}=0,$$

subject to the boundary condition

$$u(a,\theta)=f(\theta)$$

Let $u(r,\theta)=\phi(\theta)\,G(r)$. We require periodic boundary conditions in θ that ensure continuity of $\phi(\theta)$; they are

$$\phi(-\pi)=\phi(\pi)$$
$$\frac{d\phi}{d\theta}(-\pi)=\frac{d\phi}{d\theta}(\pi)$$

Now substitute $\phi(\theta)\,G(r)$ into Laplace's equation:

$$\nabla^2u=\frac{\phi(\theta)}{r}\frac{d}{dr}\left(r\frac{dG}{dr}\right)+\frac{G(r)}{r^2}\frac{d^2\phi}{d\theta^2}=0$$

Divide through by $\frac{1}{r^2}\phi(\theta)\,G(r)$ and subtract the θ equation to get

$$\frac{r}{G(r)}\frac{d}{dr}\left(r\frac{dG}{dr}\right)=-\frac{1}{\phi(\theta)}\frac{d^2\phi}{d\theta^2}=\lambda$$

where, since we require oscillation in θ to satisfy the periodic boundary conditions, we set equal to λ. So let's solve that eigenvalue problem; its solution is (assuming $\lambda > 0$)

$$\phi(\theta) = c_1 \cos\left(\sqrt{\lambda}\theta\right) + c_2 \sin\left(\sqrt{\lambda}\theta\right)$$

Imposing the boundary conditions produces

$$\begin{aligned}
\phi(-\pi) &= c_1 \cos\left(\sqrt{\lambda}\pi\right) - c_2 \sin\left(\sqrt{\lambda}\pi\right) \\
\phi(\pi) &= c_1 \cos\left(\sqrt{\lambda}\pi\right) + c_2 \sin\left(\sqrt{\lambda}\pi\right) \\
\implies 0 &= 2c_2 \sin\left(\sqrt{\lambda}\pi\right) \\
\frac{d\phi}{d\theta}(-\pi) &= c_1\sqrt{\lambda} \sin\left(\sqrt{\lambda}\pi\right) + c_2\sqrt{\lambda} \cos\left(\sqrt{\lambda}\pi\right) \\
\frac{d\phi}{d\theta}(\pi) &= -c_1\sqrt{\lambda} \sin\left(\sqrt{\lambda}\pi\right) + c_2\sqrt{\lambda} \cos\left(\sqrt{\lambda}\pi\right) \\
\implies 0 &= c_1\sqrt{\lambda} \sin\left(\sqrt{\lambda}\pi\right)
\end{aligned}$$

where a nontrivial solution exists when

$$\lambda_n = n^2, \quad n = 1, 2, \ldots$$

Thus our eigenfunctions are

$$\phi(\theta) = \sin(n\theta)\,, \cos(n\theta)$$

Next we solve the r equation (first assuming $\lambda > 0$), which we now write as

$$\frac{r}{G}\frac{d}{dr}\left(r\frac{dG}{dr}\right) = n^2 \qquad \implies \qquad r^2\frac{d^2G}{dr^2} + r\frac{dG}{dr} - n^2G = 0$$

to make its equidimensional form more apparent. One more boundedness condition is needed: require $|u(0,\theta)| < \infty$, and particularly, $|G(0)| < \infty$. In other words, u must be finite at the origin. Let $G(r) = r^p$. Solve as

$$p(p-1)\,r^p + pr^p - n^2r^p = 0 \qquad \implies \qquad p^2 - n^2 = 0 \qquad \implies \qquad p = \pm n$$

which leaves us with solution

$$G(r) = a_1 r^n + a_2 r^{-n}$$

Applying the boundedness condition, we can conclude that $a_2 = 0$ and eigenfunction

$$G(r) = r^n$$

In the second case where $\lambda = 0$, we endeavor to solve

$$\frac{r}{G}\frac{d}{dr}\left(r\frac{dG}{dr}\right) = 0$$

Dividing by r/G and integrating twice, derive solution

$$G(r) = b_1 + b_2 \ln(r)$$

and applying the boundedness condition once more forces $b_2 = 0$, leaving the constant eigenfunction.

Taking the product of the eigenfunctions and assembling their linear combination, we have solution

$$u(r,\theta) = C + \sum_{n=1}^{\infty} A_n r^n \cos(n\theta) + \sum_{n=1}^{\infty} B_n r^n \sin(n\theta)\,, \quad -\pi \le \theta \le \pi$$

For our last (non-homogeneous!) boundary condition, set $r = a$ and write

$$u(a,\theta) = C + \sum_{n=1}^{\infty} A_n a^n \cos(n\theta) + \sum_{n=1}^{\infty} B_n a^n \sin(n\theta) = f(\theta)$$

Multiply through first by $\cos(m\theta)$, integrate. Repeat with $\sin(m\theta)$ and integrate. Lastly, for the constant term, no multiplication – simply integrate over $[-\pi, \pi]$. Orthogonality produces for us the coefficients

$$\begin{aligned} C &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta)\ d\theta \\ A_n a^n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(\theta) \cos(n\theta)\ d\theta \\ B_n a^n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(\theta) \sin(n\theta)\ d\theta \end{aligned}$$